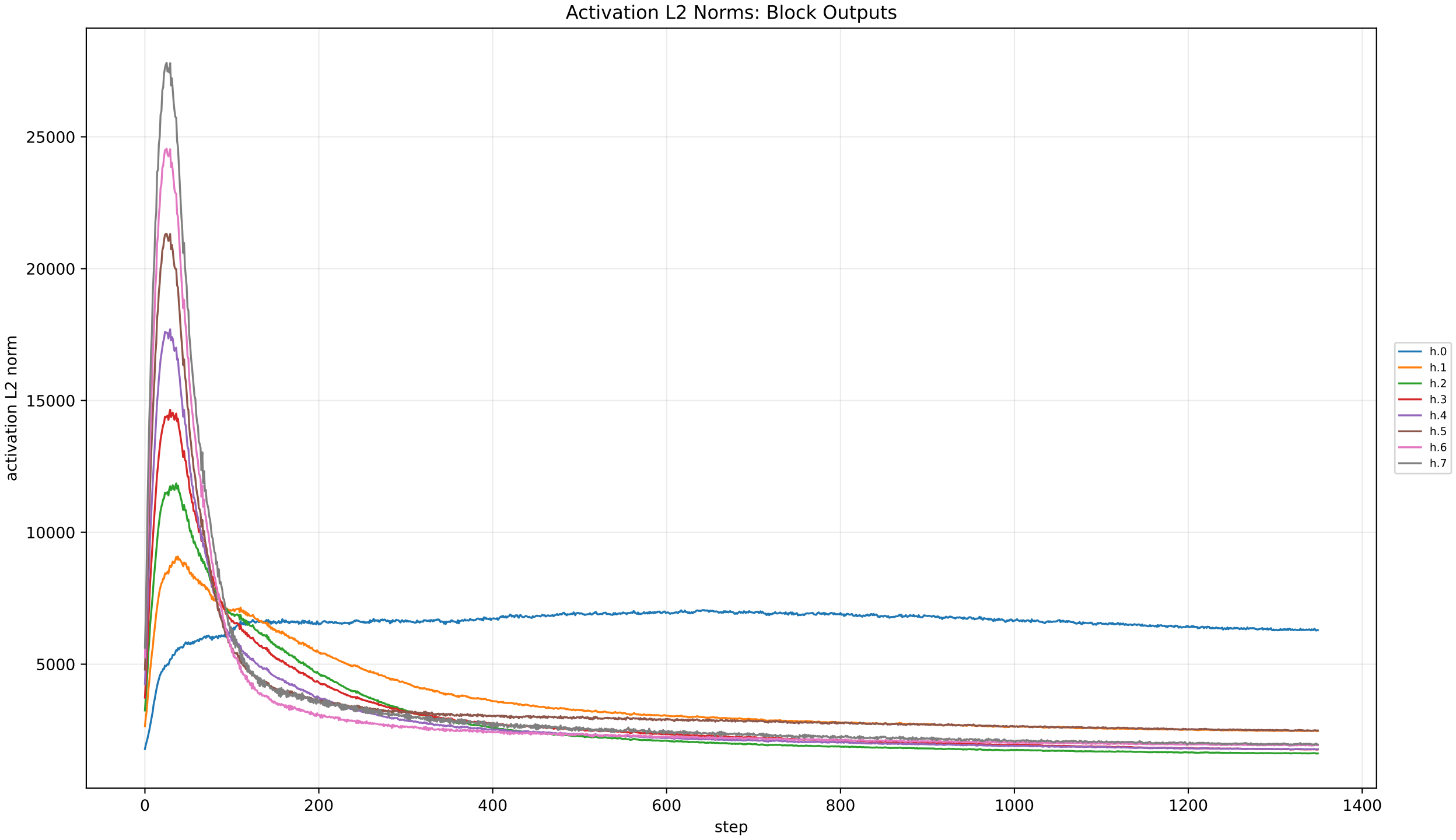
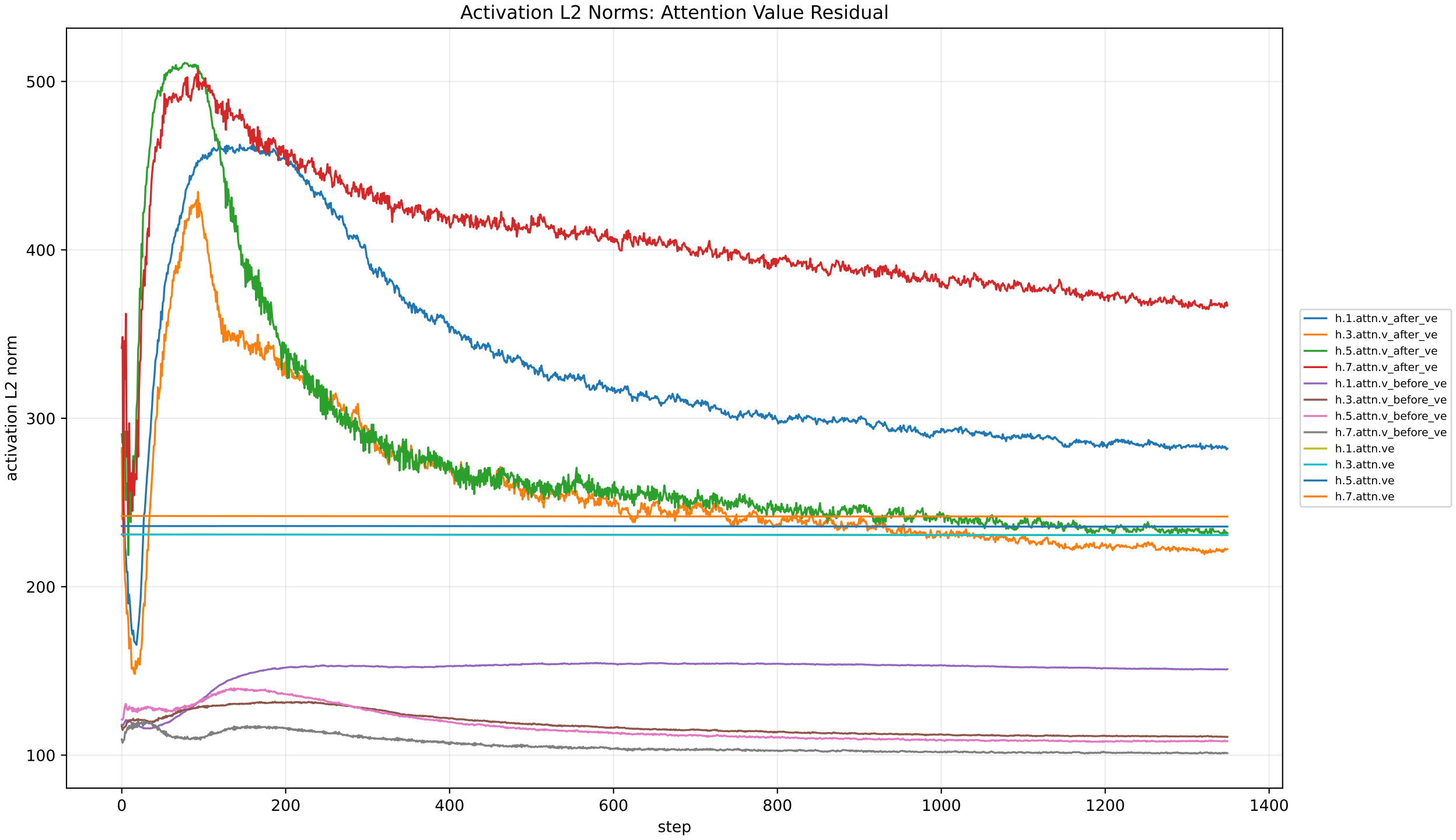
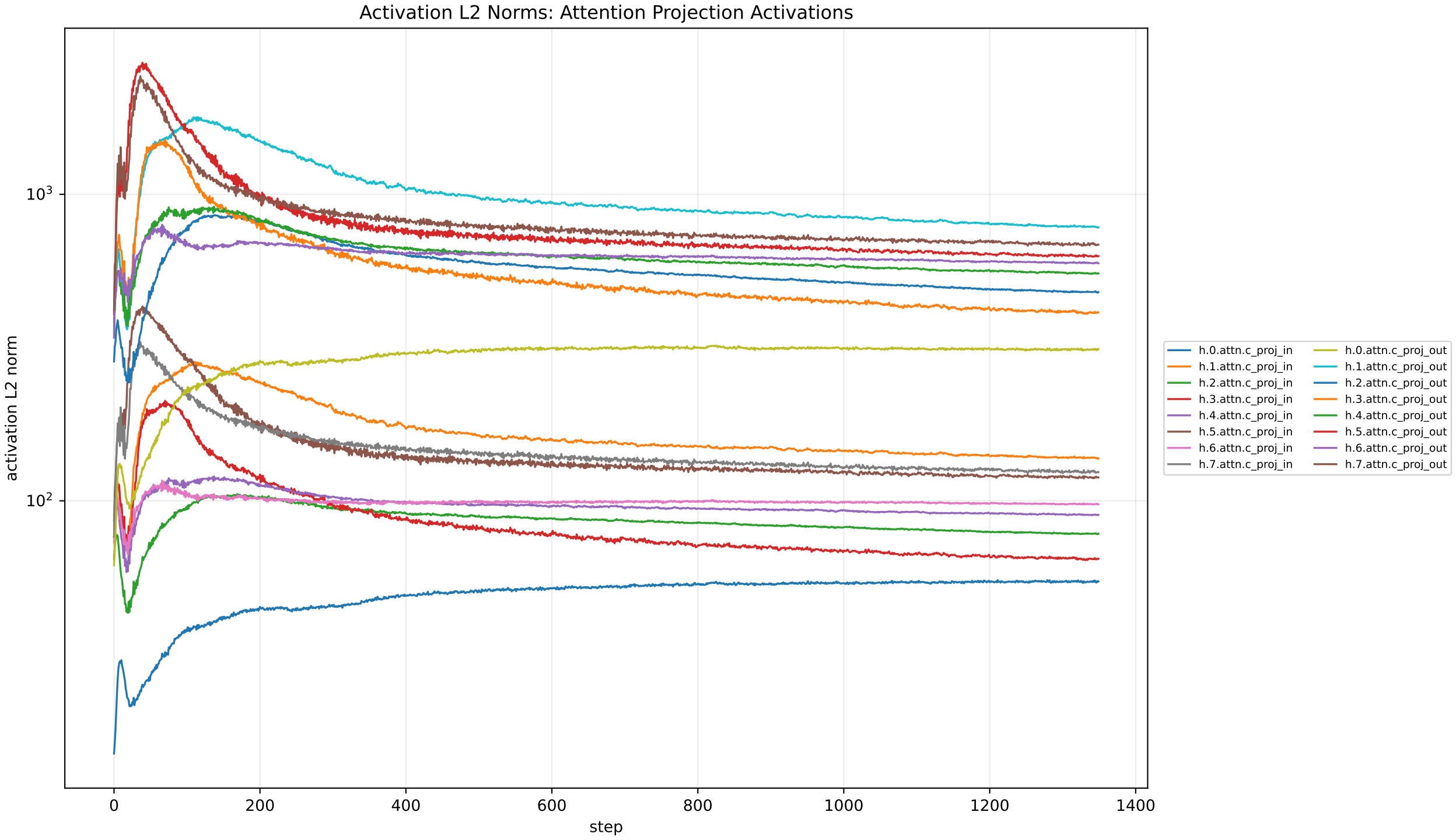
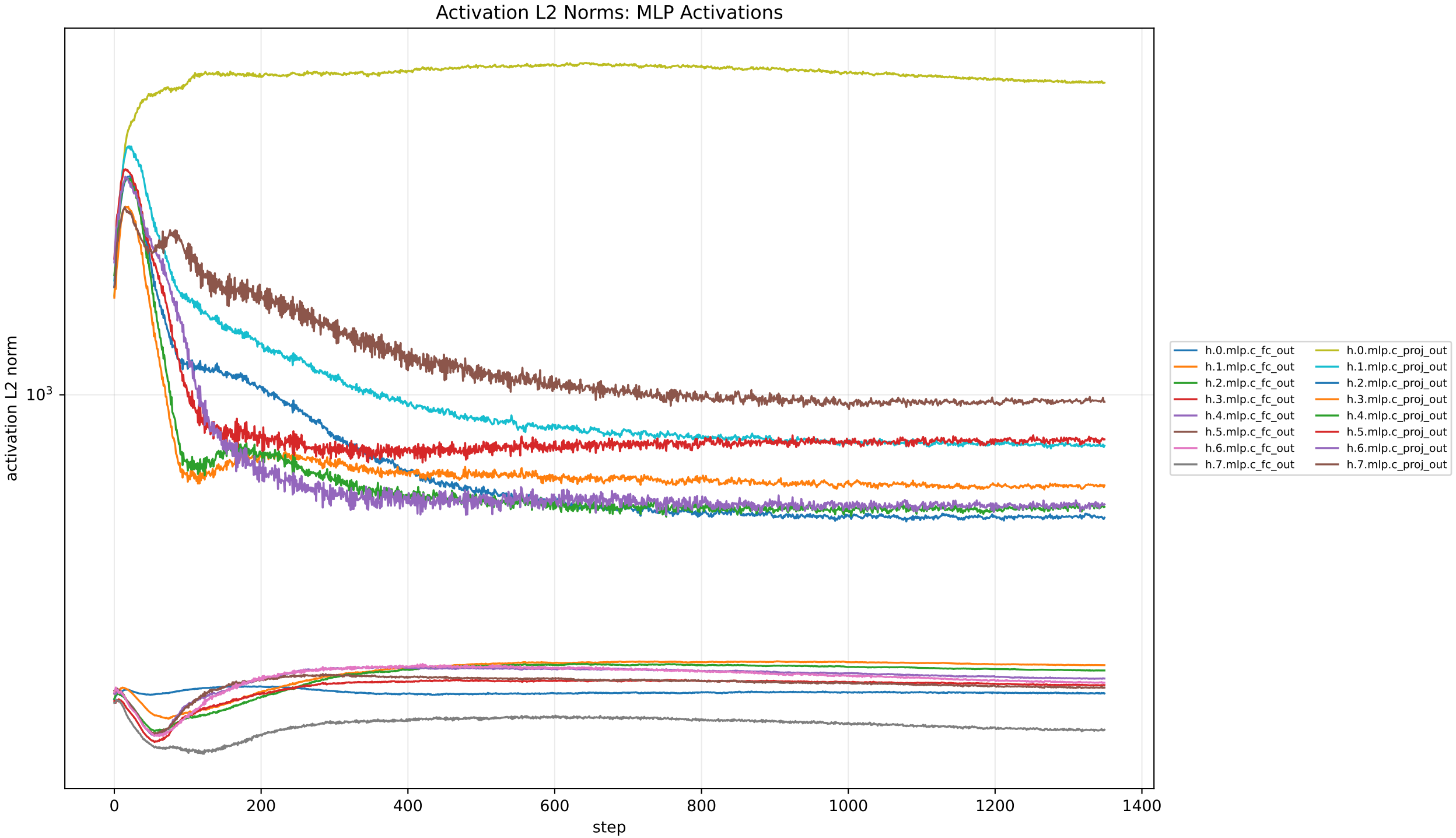
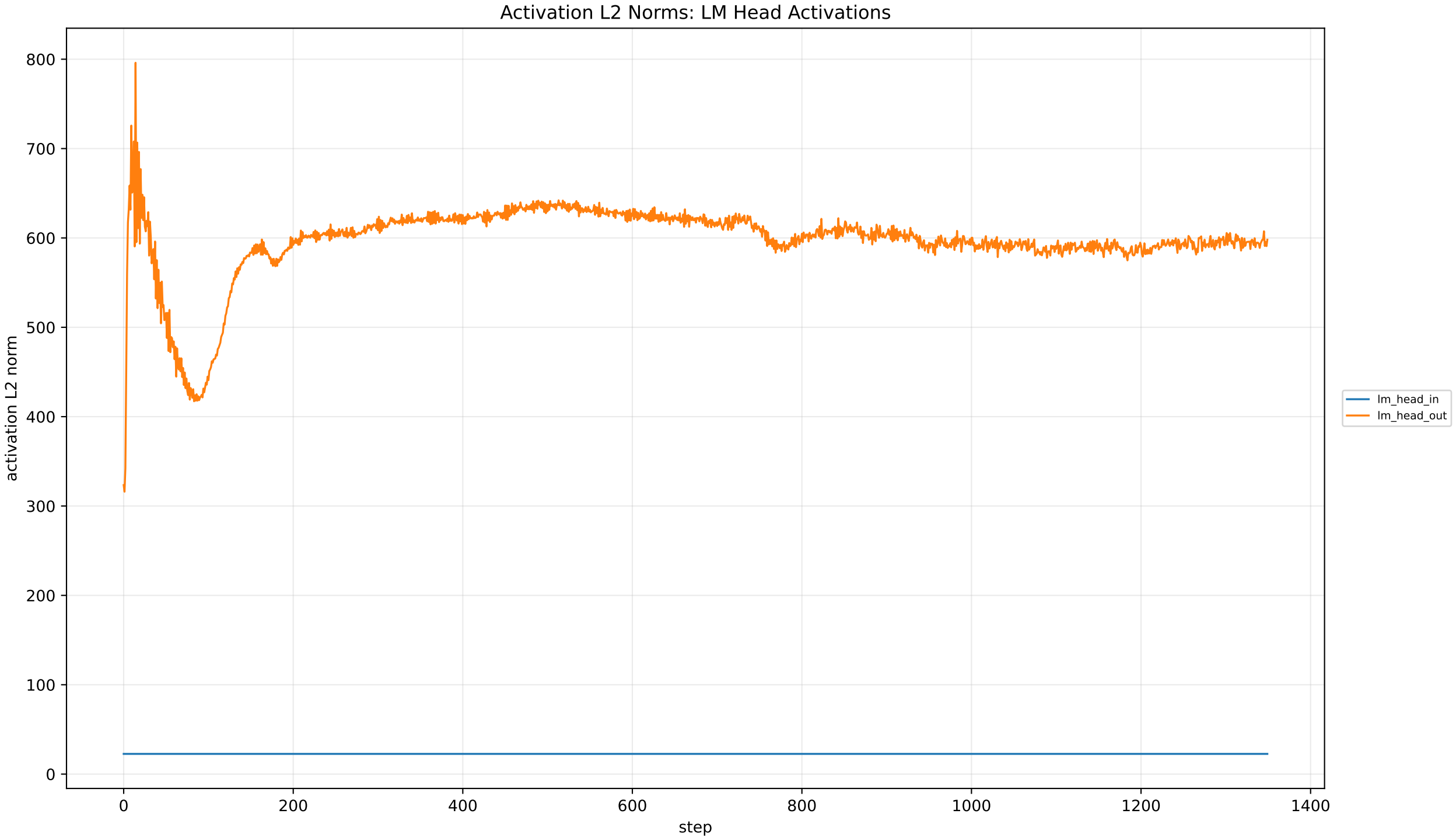
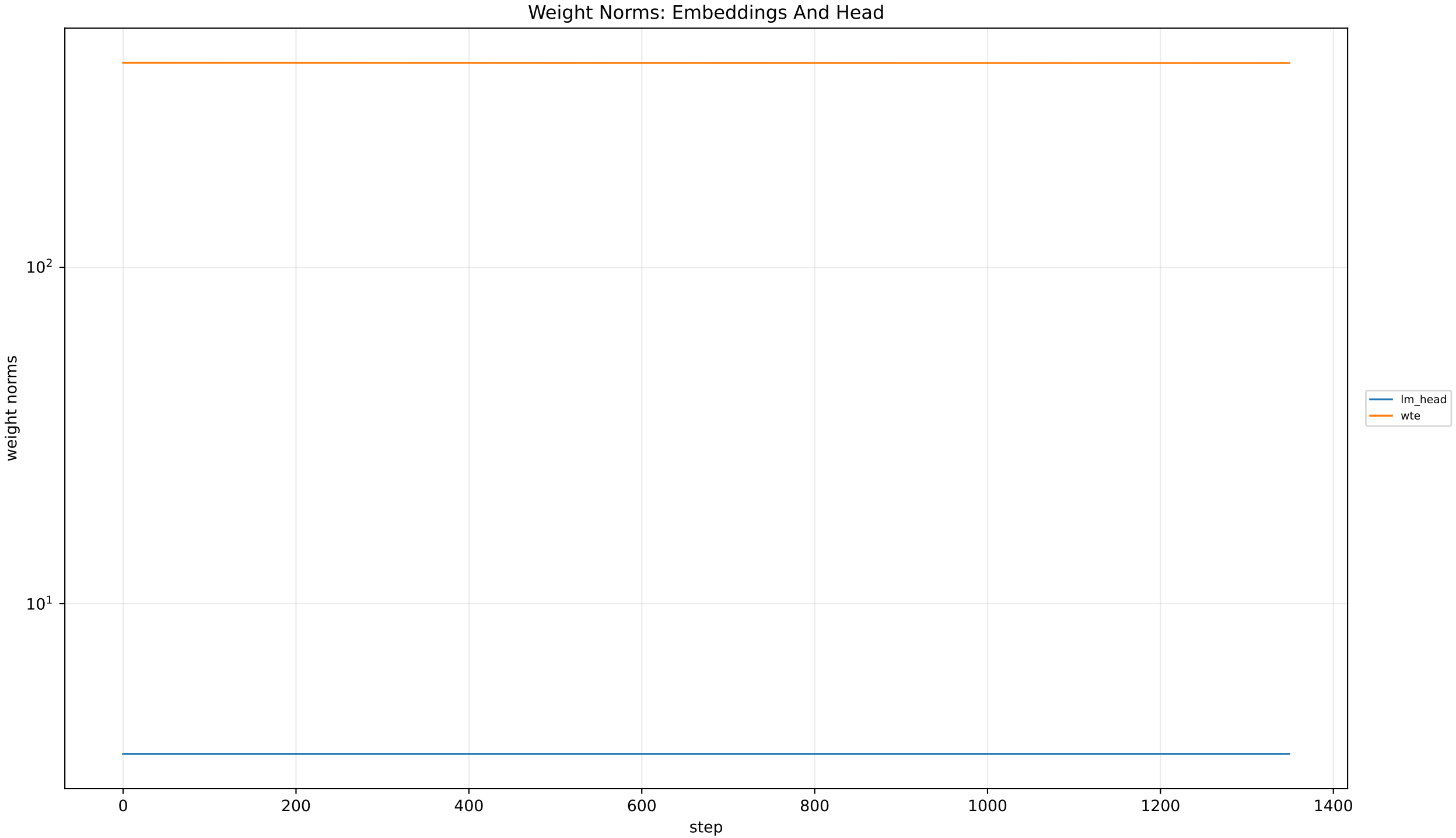

Activation L2 Norms: Block Outputs
activation L2 norm
25000
20000
15000
10000
5000
0
200
400
600
800
1000
1200
1400
step
h.0
h.1
h.2
h.3
h.4
h.5
h.6
h.7

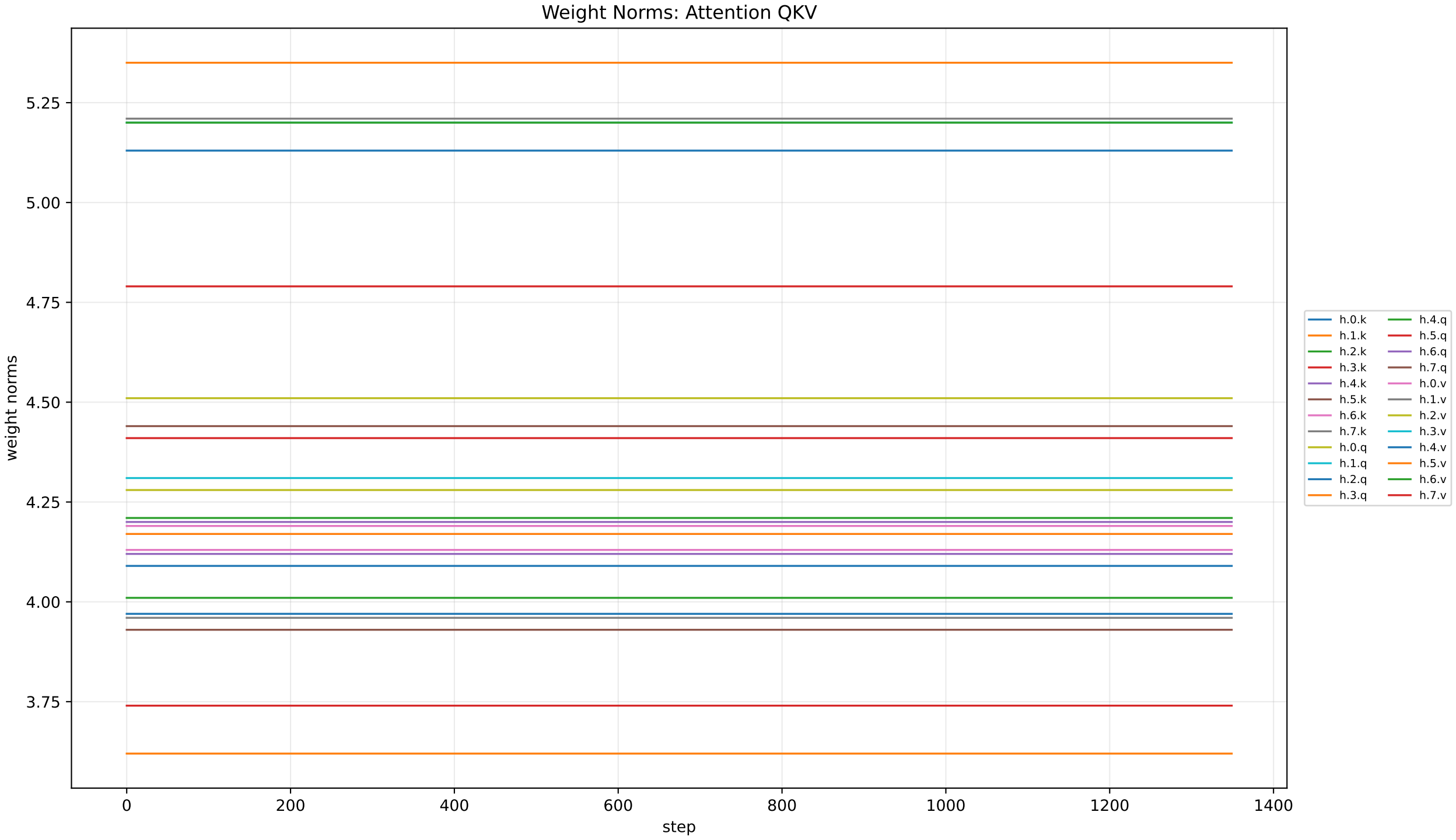

Activation L2 Norms: Attention Value Residual
activation L2 norm
step
500
400
300
200
100
0
200
400
600
800
1000
1200
1400
h.1.attn.v_after_ve
h.3.attn.v_after_ve
h.5.attn.v_after_ve
h.7.attn.v_after_ve
h.1.attn.v_before_ve
h.3.attn.v_before_ve
h.5.attn.v_before_ve
h.7.attn.v_before_ve
h.1.attn.ve
h.3.attn.ve
h.5.attn.ve
h.7.attn.ve

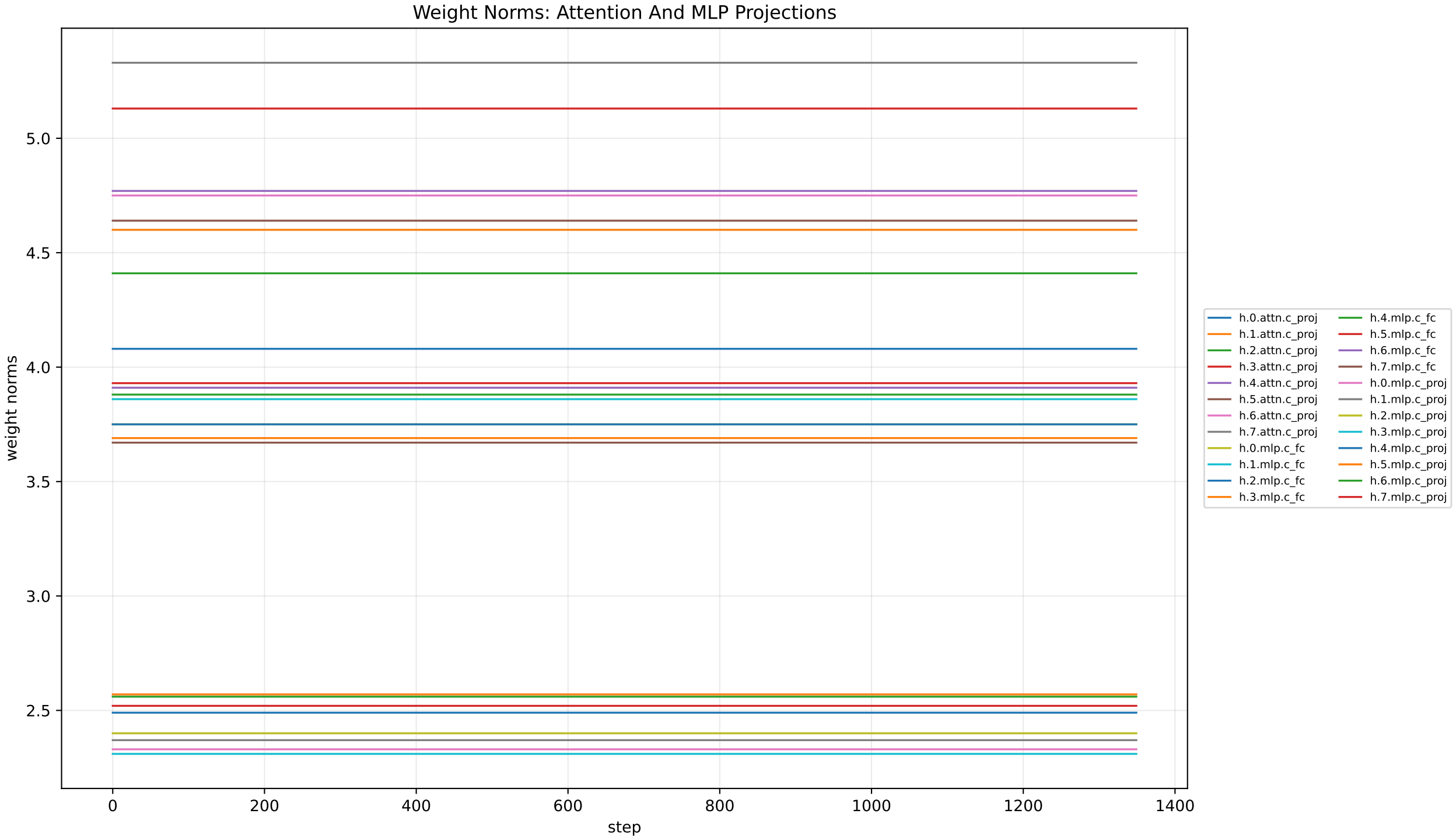

Activation L2 Norms: Attention Projection Activations
activation L2 norm
10^3
10^2
0
200
400
600
800
1000
1200
1400
step
h.0.attn.c_proj_in
h.1.attn.c_proj_in
h.2.attn.c_proj_in
h.3.attn.c_proj_in
h.4.attn.c_proj_in
h.5.attn.c_proj_in
h.6.attn.c_proj_in
h.7.attn.c_proj_in
h.0.attn.c_proj_out
h.1.attn.c_proj_out
h.2.attn.c_proj_out
h.3.attn.c_proj_out
h.4.attn.c_proj_out
h.5.attn.c_proj_out
h.6.attn.c_proj_out
h.7.attn.c_proj_out

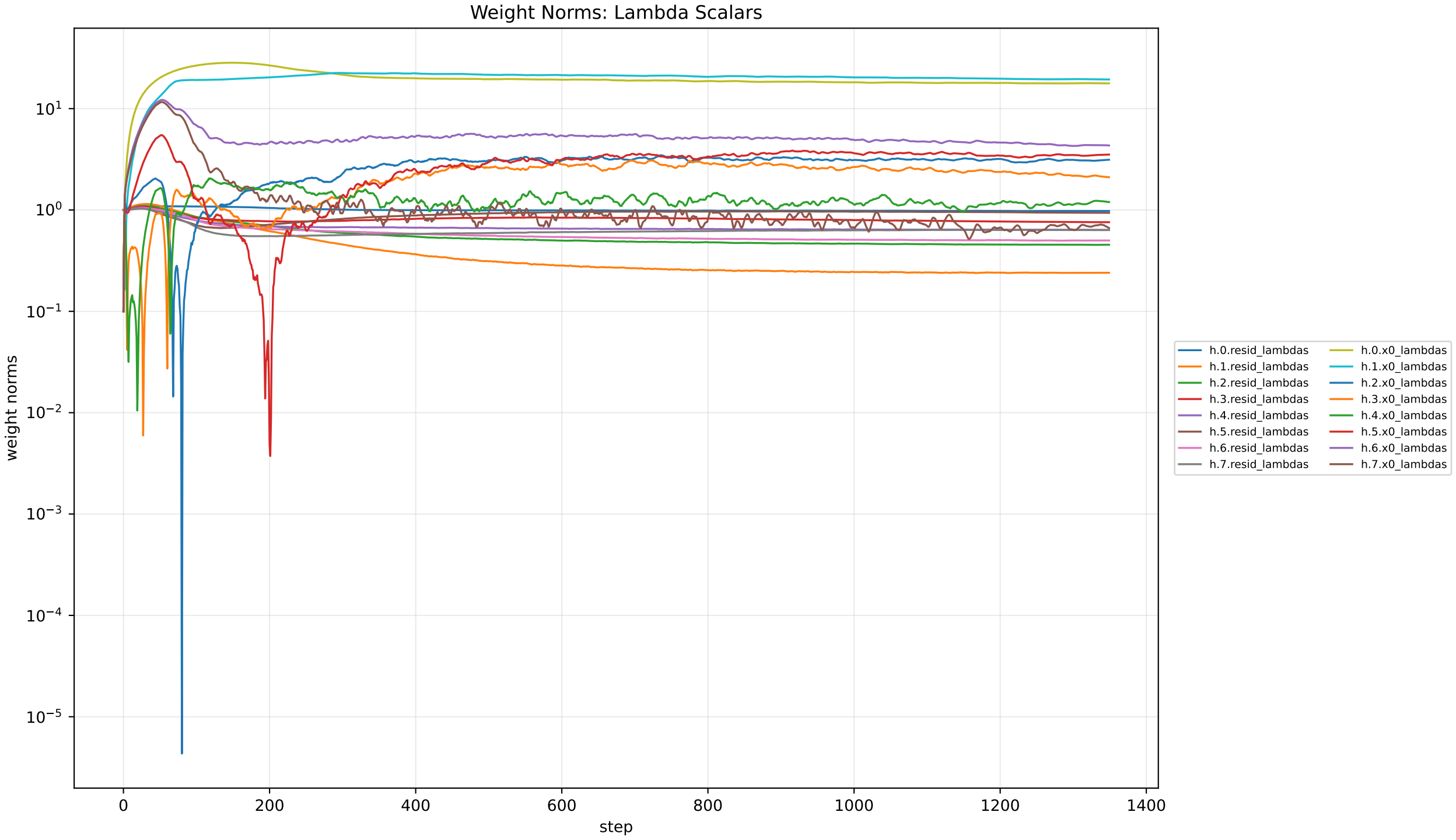

Activation L2 Norms: MLP Activations
activation L2 norm
10^3
0
200
400
600
800
1000
1200
1400
step
h.0.mlp.c_fc_out
h.1.mlp.c_fc_out
h.2.mlp.c_fc_out
h.3.mlp.c_fc_out
h.4.mlp.c_fc_out
h.5.mlp.c_fc_out
h.6.mlp.c_fc_out
h.7.mlp.c_fc_out
h.0.mlp.c_proj_out
h.1.mlp.c_proj_out
h.2.mlp.c_proj_out
h.3.mlp.c_proj_out
h.4.mlp.c_proj_out
h.5.mlp.c_proj_out
h.6.mlp.c_proj_out
h.7.mlp.c_proj_out

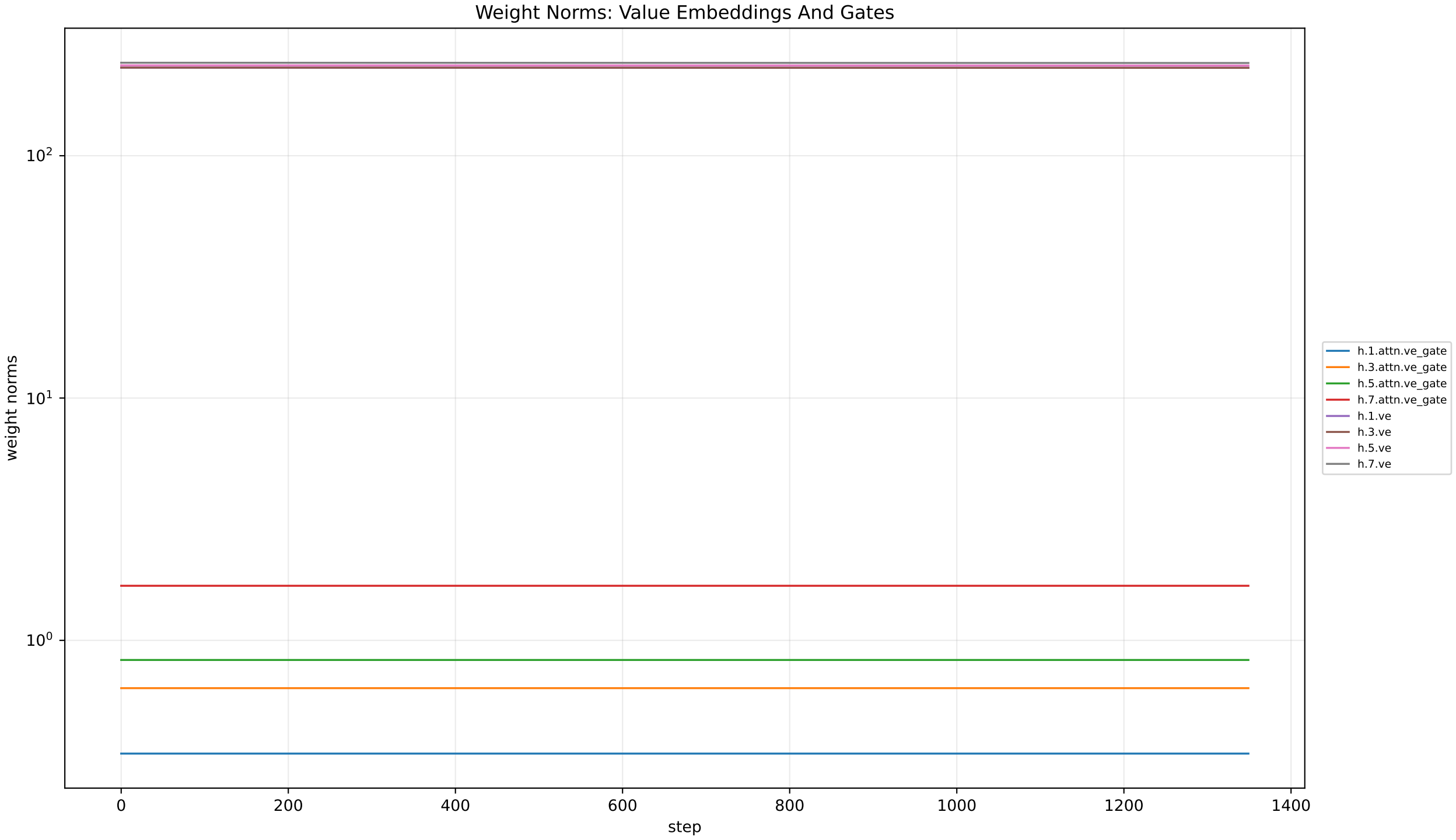

Activation L2 Norms: LM Head Activations
activation L2 norm
step
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
1400
lm_head_in
lm_head_out

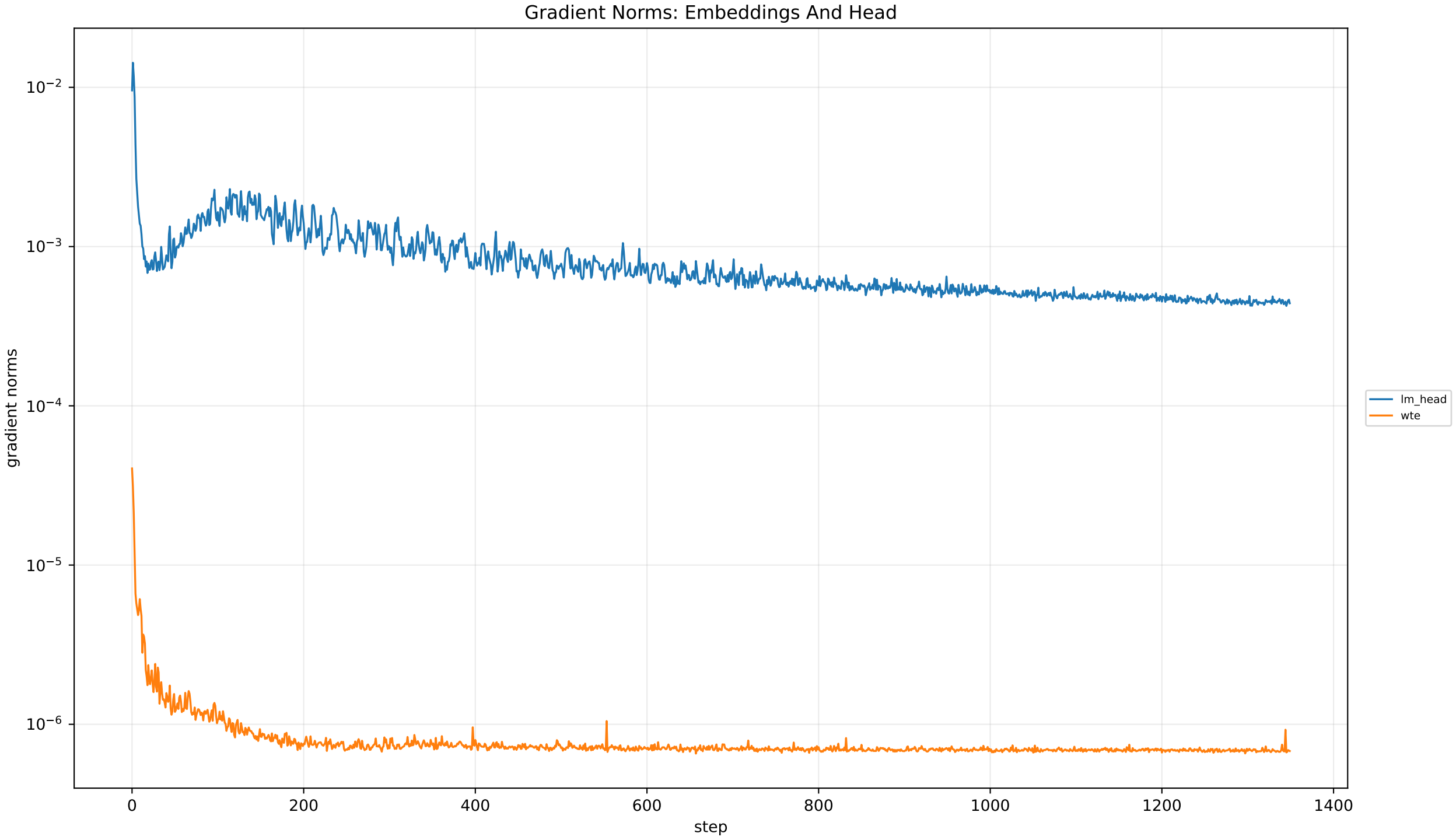

Weight Norms: Embeddings And Head
weight norms
10^2
10^1
0
200
400
600
800
1000
1200
1400
step
lm_head
wte

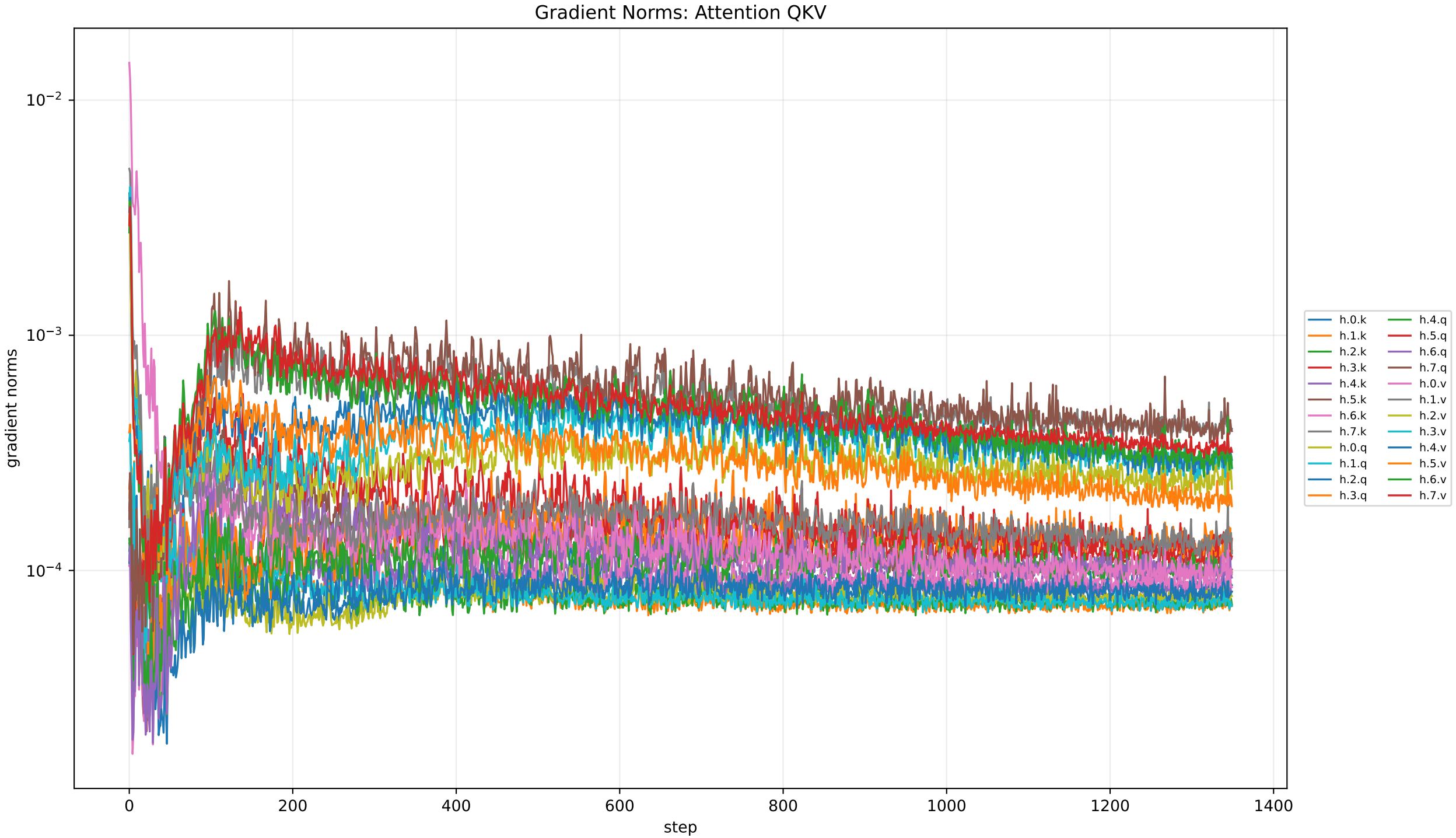

Weight Norms: Attention QKV
weight norms
5.25
5.00
4.75
4.50
4.25
4.00
3.75
0
200
400
600
800
1000
1200
1400
step
h.0.k
h.1.k
h.2.k
h.3.k
h.4.k
h.5.k
h.6.k
h.7.k
h.0.q
h.1.q
h.2.q
h.3.q
h.4.q
h.5.q
h.6.q
h.7.q
h.0.v
h.1.v
h.2.v
h.3.v
h.4.v
h.5.v
h.6.v
h.7.v

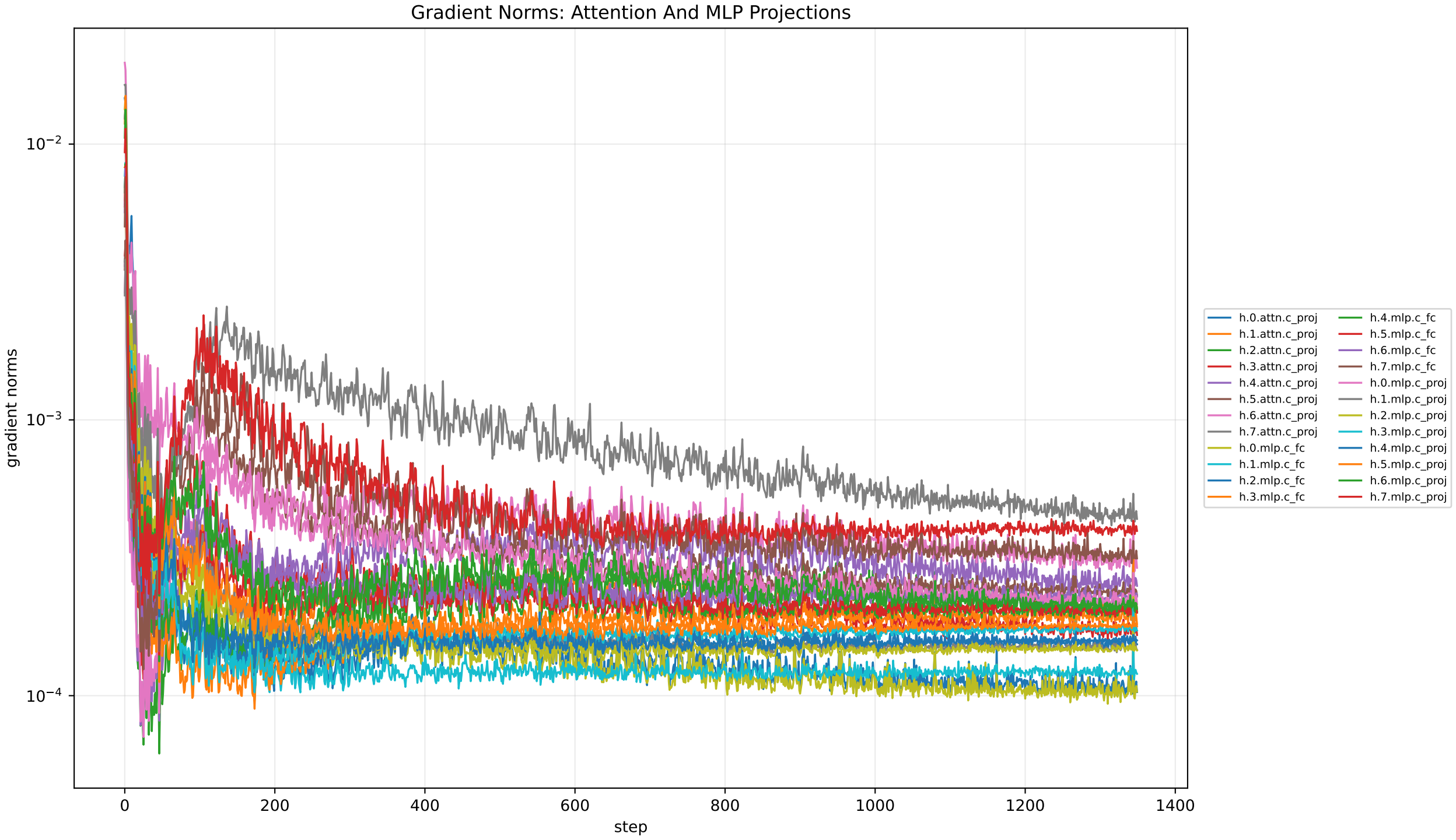

Weight Norms: Attention And MLP Projections
weight norms
step
5.0
4.5
4.0
3.5
3.0
2.5
0
200
400
600
800
1000
1200
1400
h.0.attn.c_proj
h.1.attn.c_proj
h.2.attn.c_proj
h.3.attn.c_proj
h.4.attn.c_proj
h.5.attn.c_proj
h.6.attn.c_proj
h.7.attn.c_proj
h.0.mlp.c_fc
h.1.mlp.c_fc
h.2.mlp.c_fc
h.3.mlp.c_fc
h.4.mlp.c_fc
h.5.mlp.c_fc
h.6.mlp.c_fc
h.7.mlp.c_fc
h.0.mlp.c_proj
h.1.mlp.c_proj
h.2.mlp.c_proj
h.3.mlp.c_proj
h.4.mlp.c_proj
h.5.mlp.c_proj
h.6.mlp.c_proj
h.7.mlp.c_proj

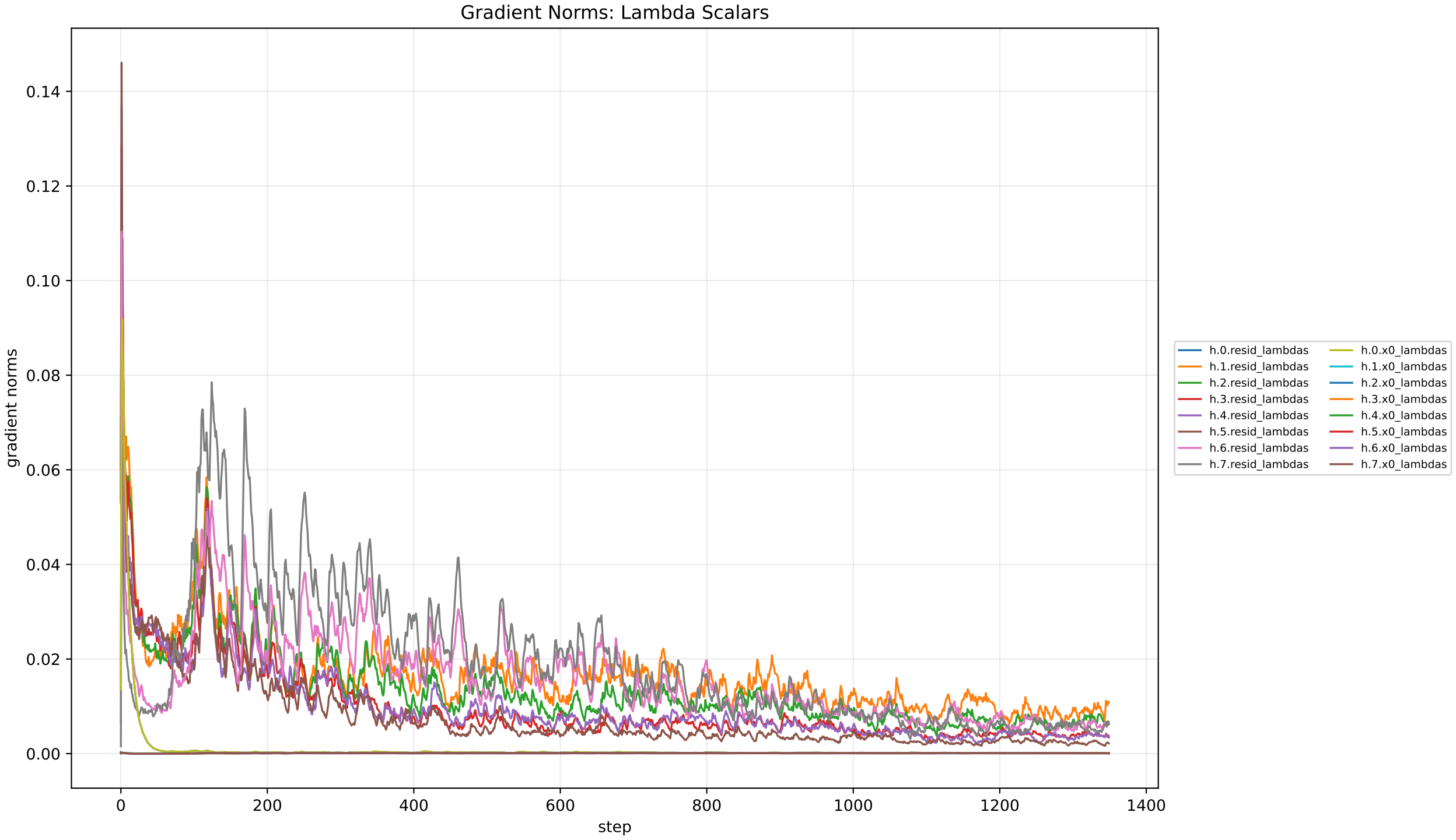

Weight Norms: Lambda Scalars
weight norms
step
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0
200
400
600
800
1000
1200
1400
h.0.resid_lambdas
h.1.resid_lambdas
h.2.resid_lambdas
h.3.resid_lambdas
h.4.resid_lambdas
h.5.resid_lambdas
h.6.resid_lambdas
h.7.resid_lambdas
h.0.x0_lambdas
h.1.x0_lambdas
h.2.x0_lambdas
h.3.x0_lambdas
h.4.x0_lambdas
h.5.x0_lambdas
h.6.x0_lambdas
h.7.x0_lambdas

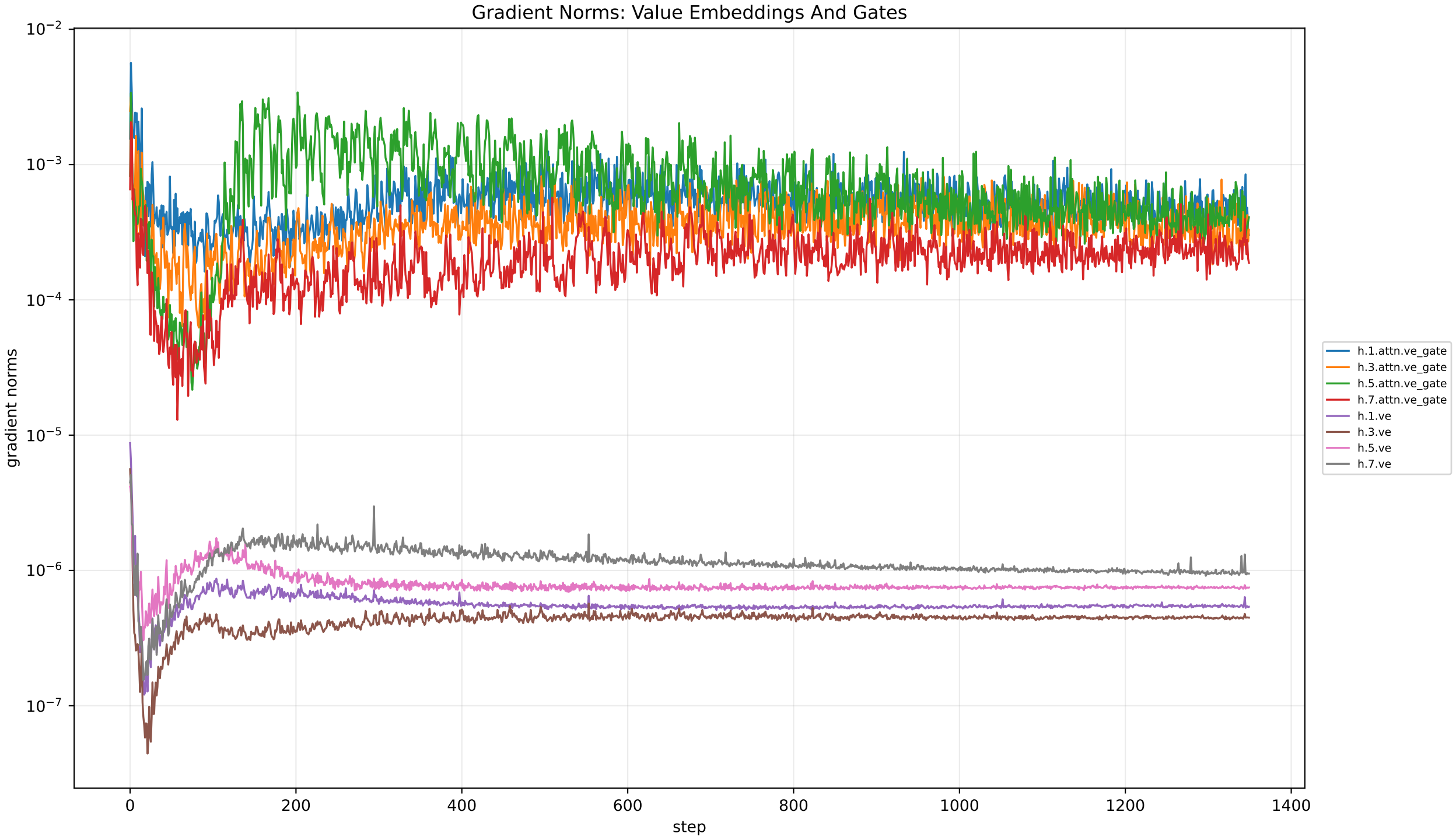

Weight Norms: Value Embeddings And Gates
weight norms
step
10^2
10^1
10^0
0
200
400
600
800
1000
1200
1400
h.1.attn.ve_gate
h.3.attn.ve_gate
h.5.attn.ve_gate
h.7.attn.ve_gate
h.1.ve
h.3.ve
h.5.ve
h.7.ve

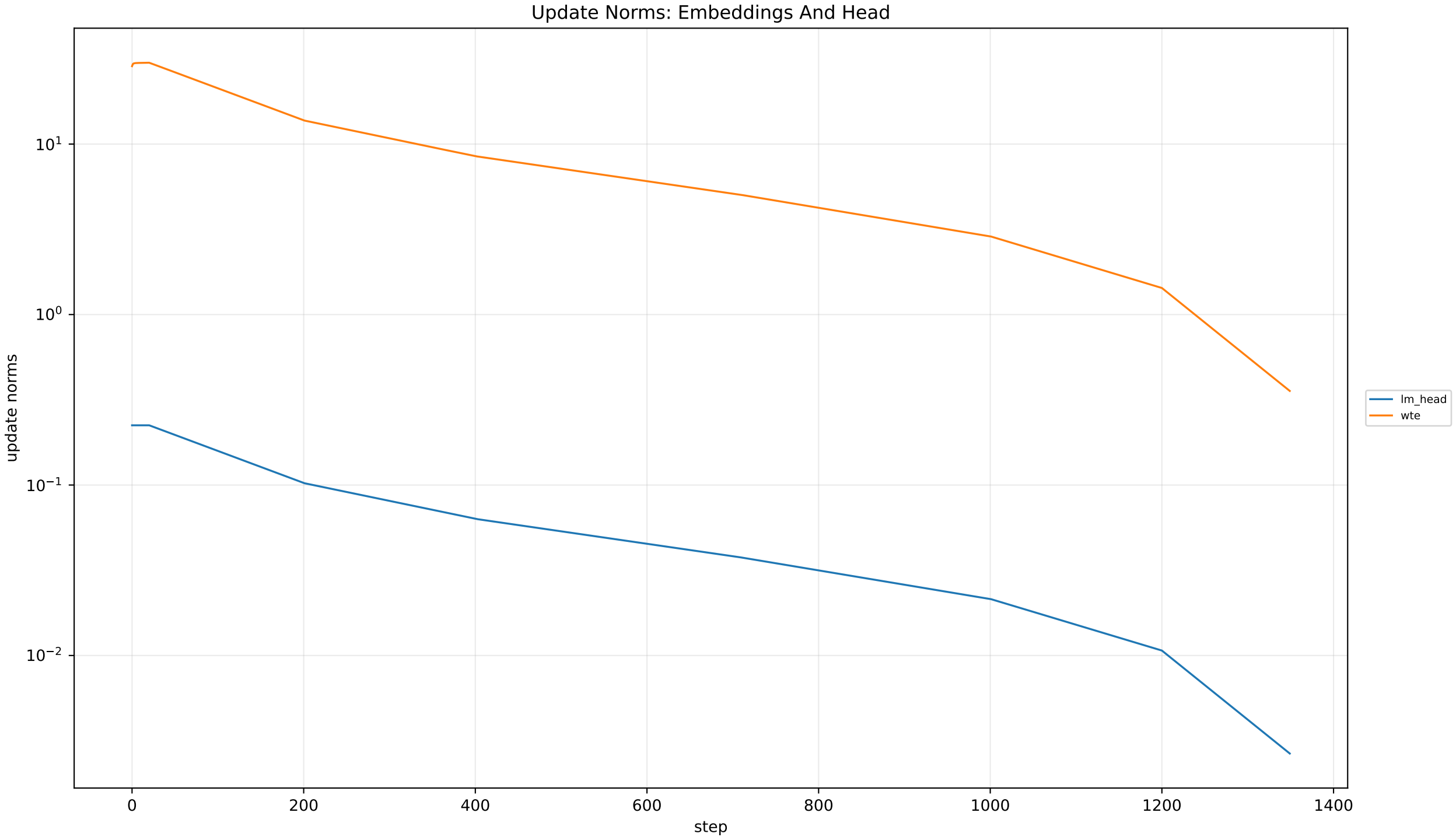

Gradient Norms: Embeddings And Head
gradient norms
step
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
0
200
400
600
800
1000
1200
1400
lm_head
wte

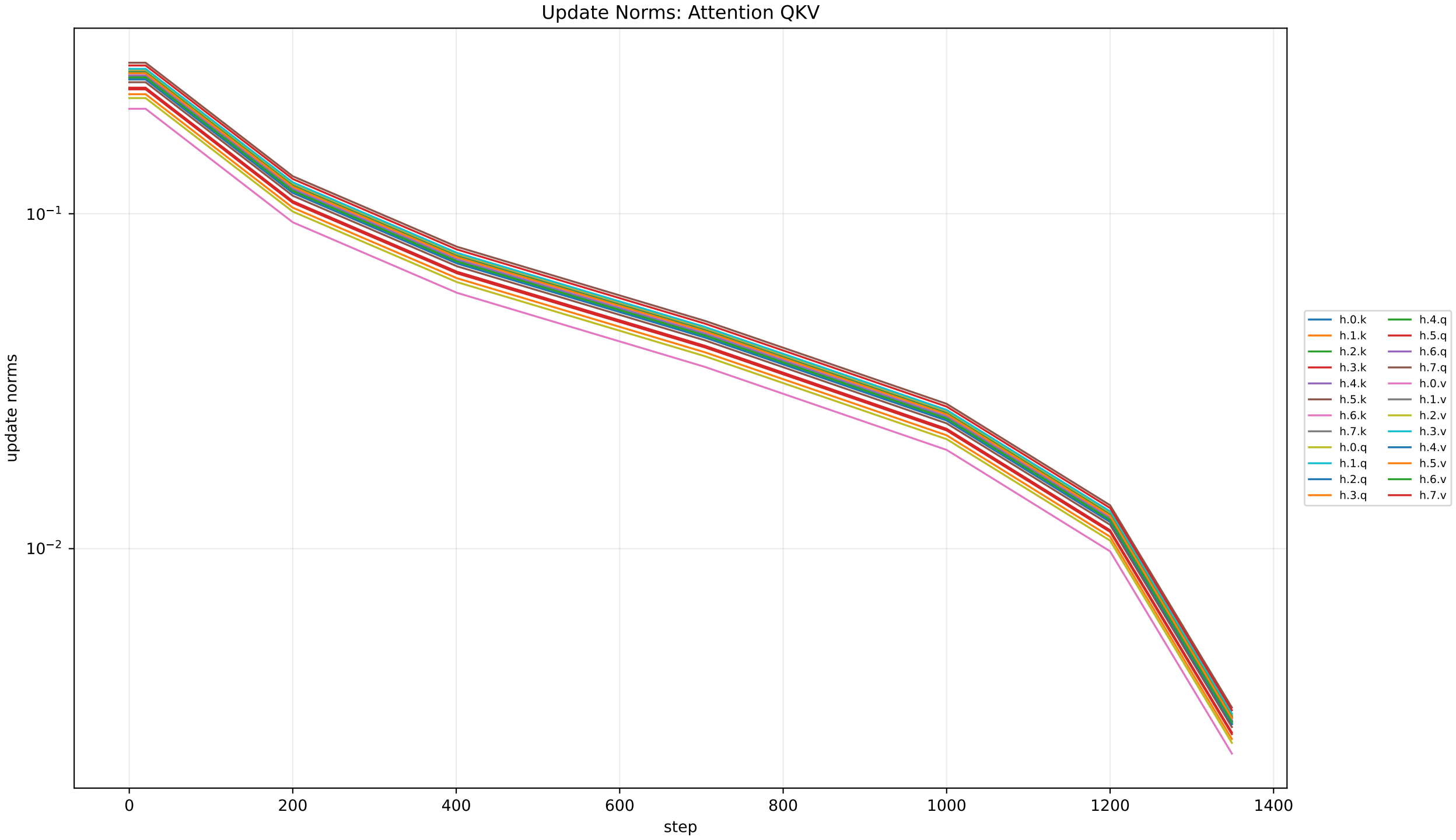

Gradient Norms: Attention QKV
gradient norms
step
10^{-2}
10^{-3}
10^{-4}
0
200
400
600
800
1000
1200
1400
h.0.k
h.1.k
h.2.k
h.3.k
h.4.k
h.5.k
h.6.k
h.7.k
h.0.q
h.1.q
h.2.q
h.3.q
h.4.q
h.5.q
h.6.q
h.7.q
h.0.v
h.1.v
h.2.v
h.3.v
h.4.v
h.5.v
h.6.v
h.7.v

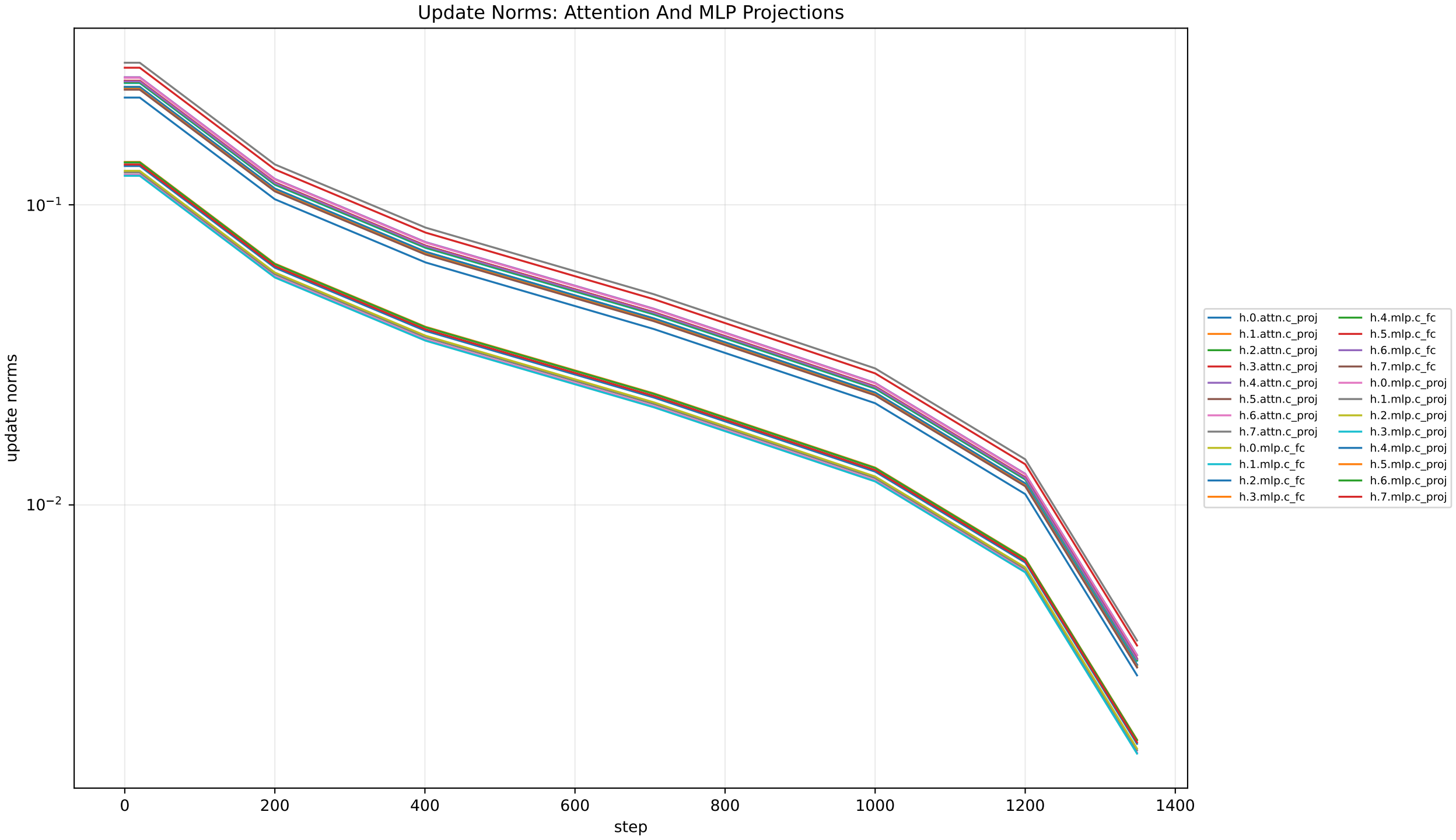

Gradient Norms: Attention And MLP Projections
gradient norms
step
10^{-2}
10^{-3}
10^{-4}
0
200
400
600
800
1000
1200
1400
h.0.attn.c_proj
h.1.attn.c_proj
h.2.attn.c_proj
h.3.attn.c_proj
h.4.attn.c_proj
h.5.attn.c_proj
h.6.attn.c_proj
h.7.attn.c_proj
h.0.mlp.c_fc
h.1.mlp.c_fc
h.2.mlp.c_fc
h.3.mlp.c_fc
h.4.mlp.c_fc
h.5.mlp.c_fc
h.6.mlp.c_fc
h.7.mlp.c_fc
h.0.mlp.c_proj
h.1.mlp.c_proj
h.2.mlp.c_proj
h.3.mlp.c_proj
h.4.mlp.c_proj
h.5.mlp.c_proj
h.6.mlp.c_proj
h.7.mlp.c_proj

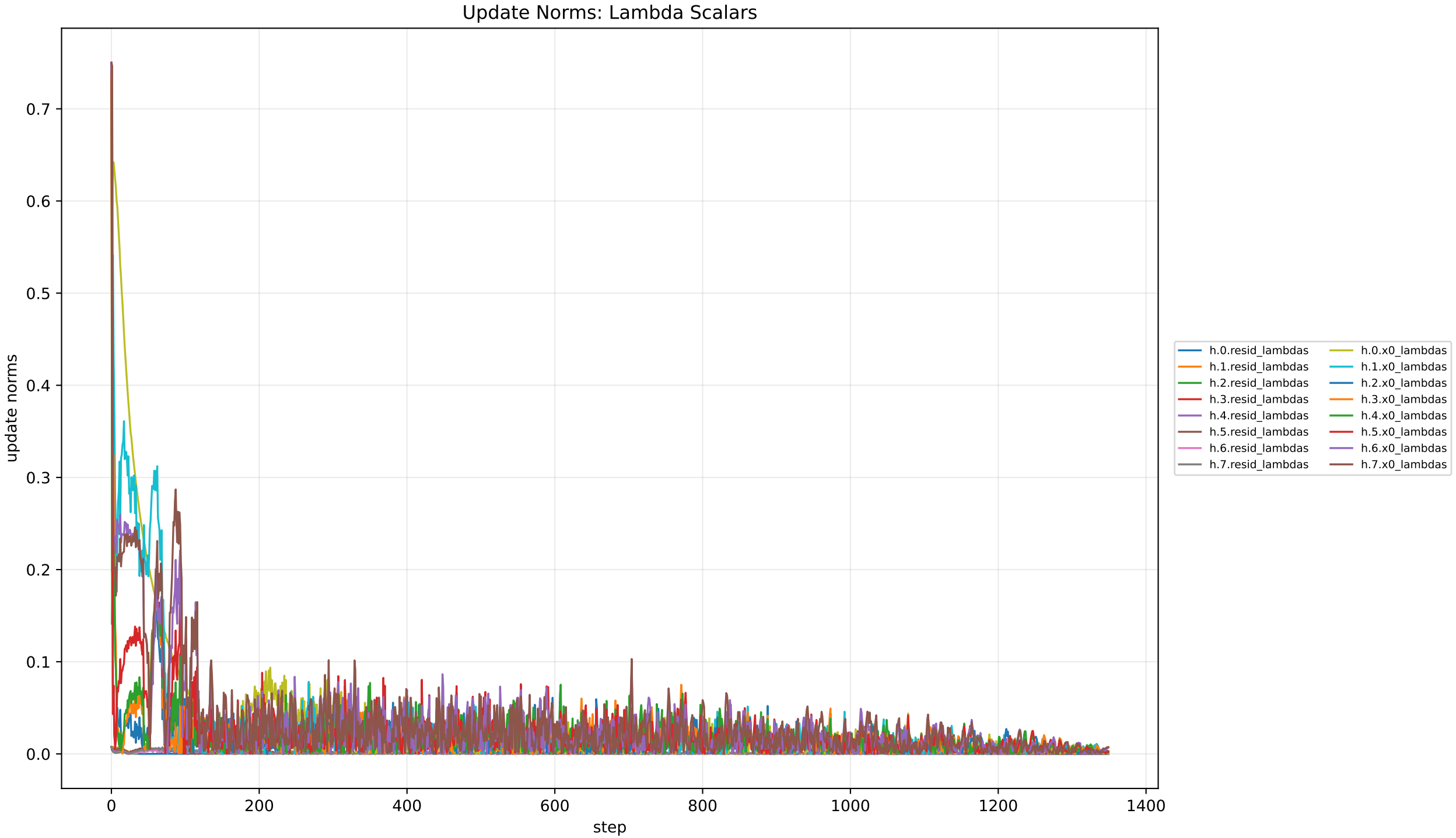

Gradient Norms: Lambda Scalars
gradient norms
step
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0
200
400
600
800
1000
1200
1400
h.0.resid_lambdas
h.1.resid_lambdas
h.2.resid_lambdas
h.3.resid_lambdas
h.4.resid_lambdas
h.5.resid_lambdas
h.6.resid_lambdas
h.7.resid_lambdas
h.0.x0_lambdas
h.1.x0_lambdas
h.2.x0_lambdas
h.3.x0_lambdas
h.4.x0_lambdas
h.5.x0_lambdas
h.6.x0_lambdas
h.7.x0_lambdas

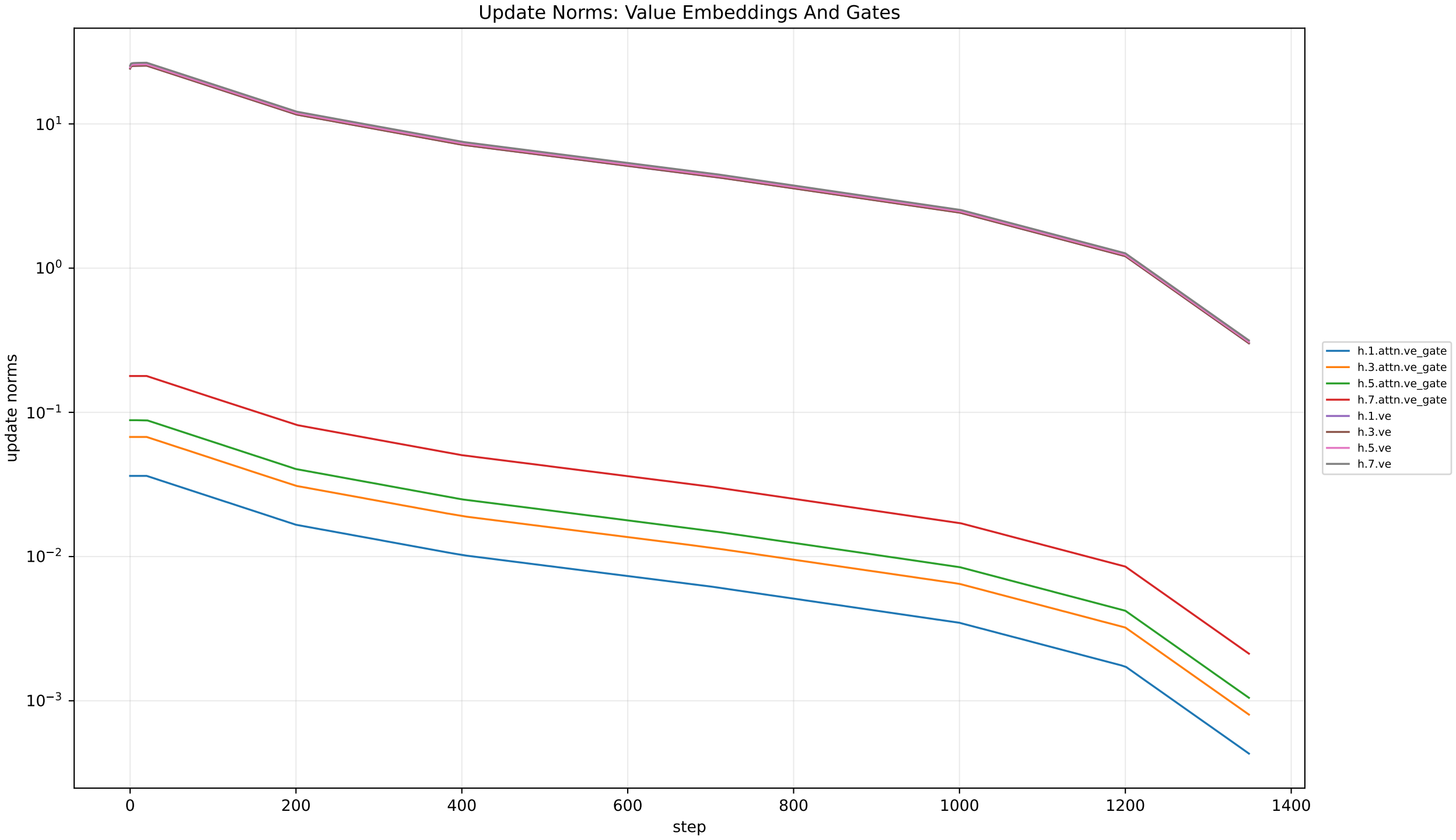

Gradient Norms: Value Embeddings And Gates
gradient norms
step
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
0
200
400
600
800
1000
1200
1400
h.1.attn.ve_gate
h.3.attn.ve_gate
h.5.attn.ve_gate
h.7.attn.ve_gate
h.1.ve
h.3.ve
h.5.ve
h.7.ve

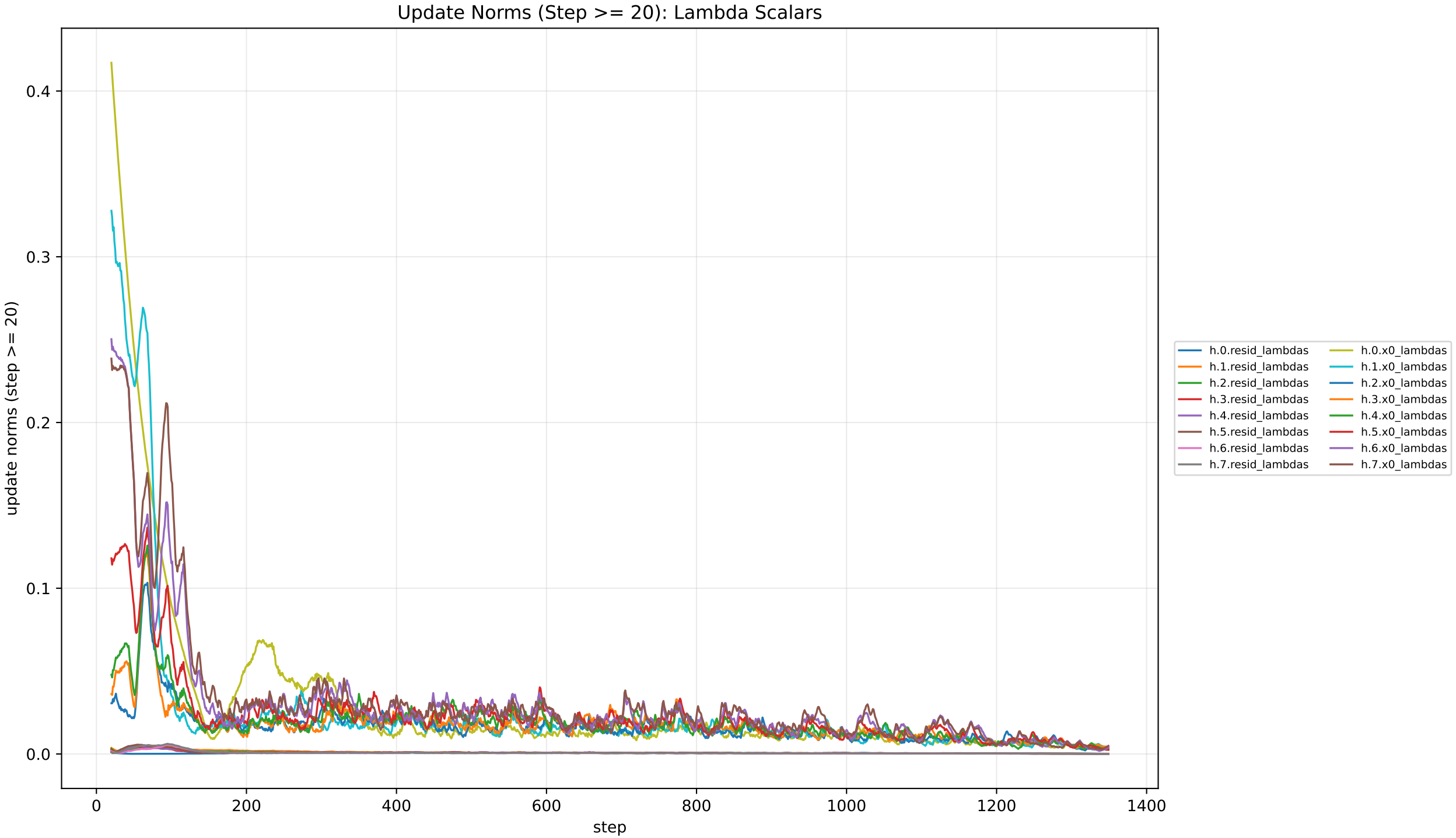

Update Norms: Embeddings And Head
update norms
step
10^{1}
10^{0}
10^{-1}
10^{-2}
0
200
400
600
800
1000
1200
1400
lm_head
wte

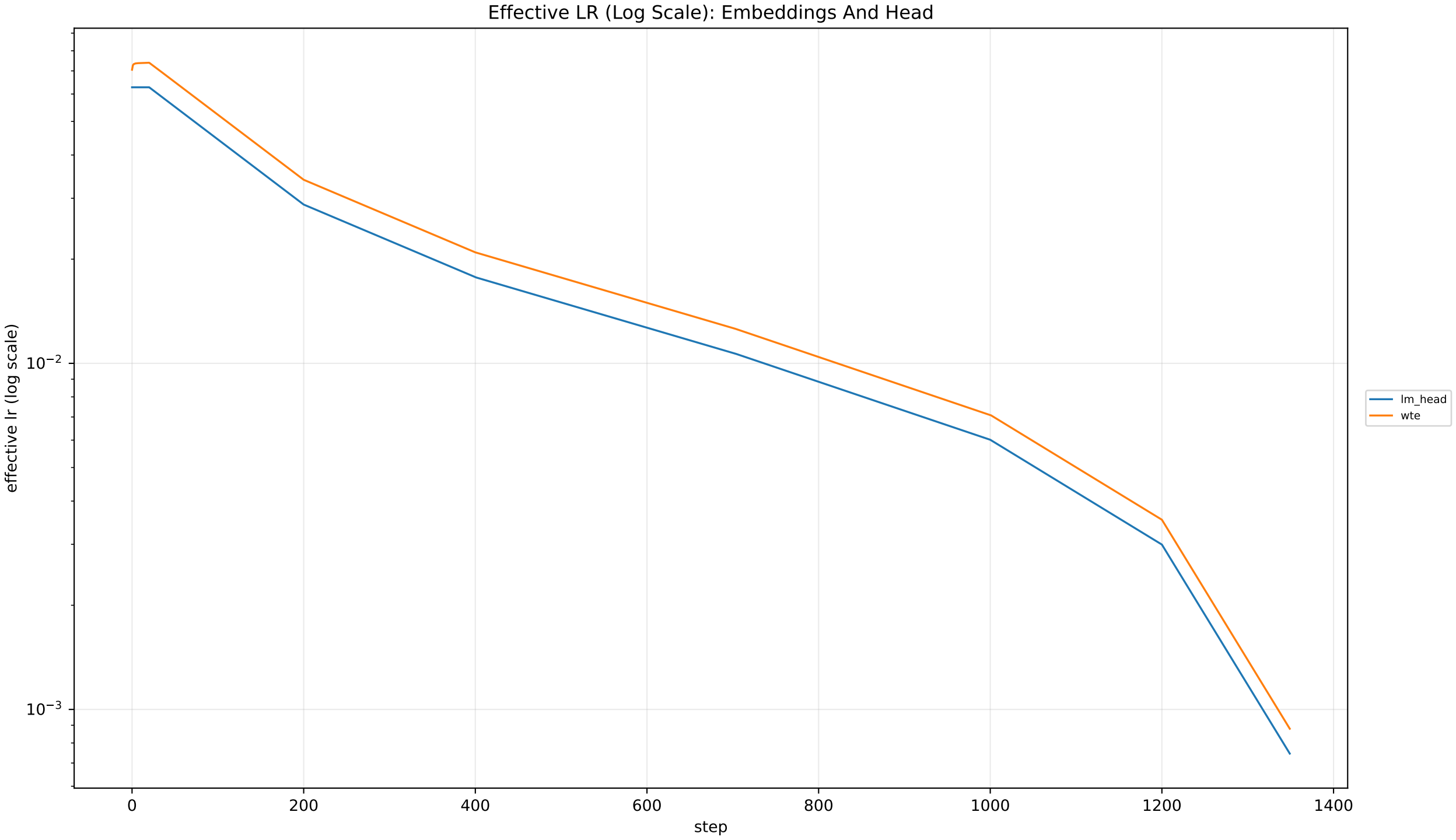

Update Norms: Attention QKV
update norms
10^{-1}
10^{-2}
step
0
200
400
600
800
1000
1200
1400
h.0.k
h.1.k
h.2.k
h.3.k
h.4.k
h.5.k
h.6.k
h.7.k
h.0.q
h.1.q
h.2.q
h.3.q
h.4.q
h.5.q
h.6.q
h.7.q
h.0.v
h.1.v
h.2.v
h.3.v
h.4.v
h.5.v
h.6.v
h.7.v

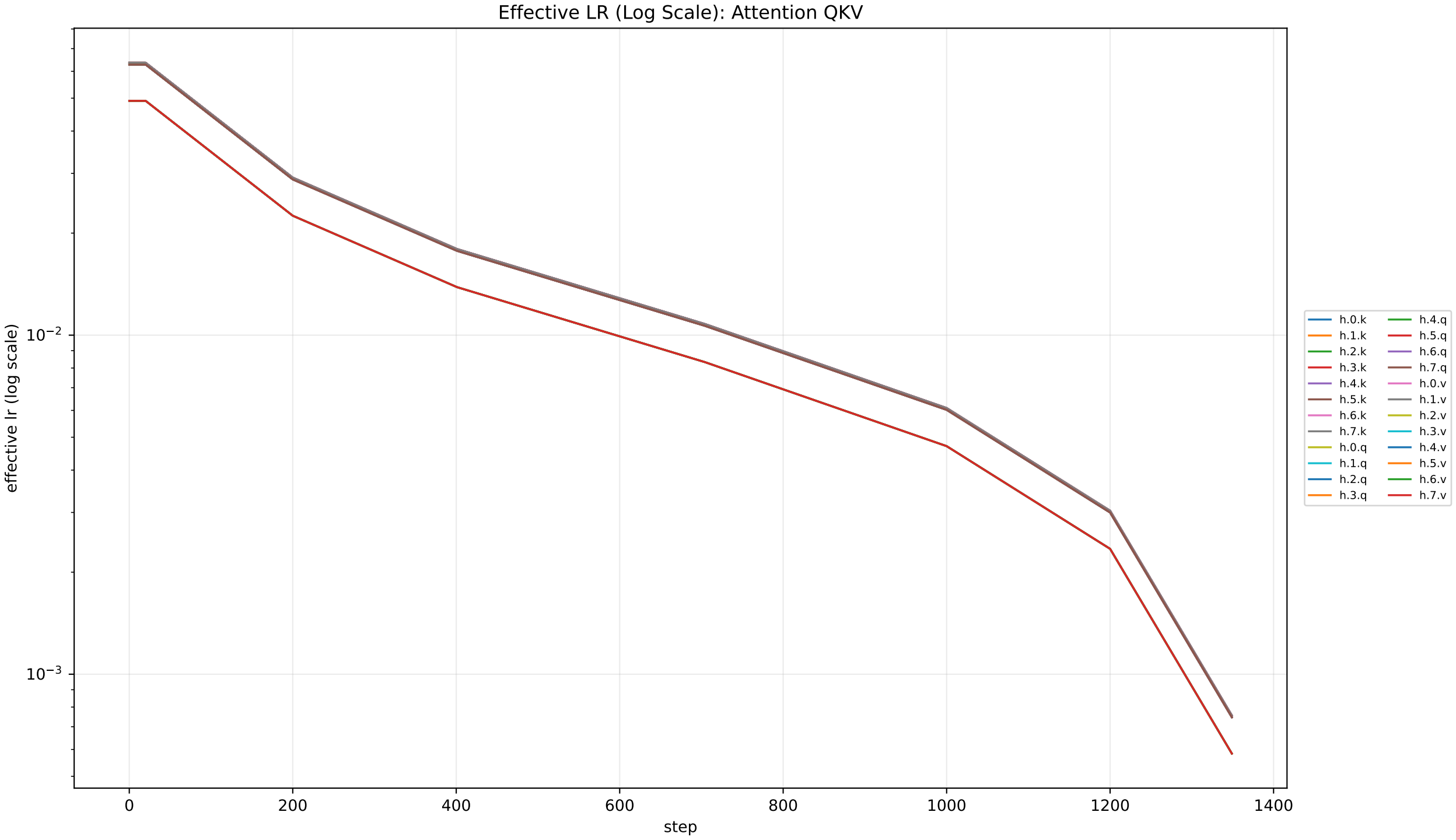

Update Norms: Attention And MLP Projections
update norms
step
10^{-1}
10^{-2}
0
200
400
600
800
1000
1200
1400
h.0.attn.c_proj
h.1.attn.c_proj
h.2.attn.c_proj
h.3.attn.c_proj
h.4.attn.c_proj
h.5.attn.c_proj
h.6.attn.c_proj
h.7.attn.c_proj
h.0.mlp.c_fc
h.1.mlp.c_fc
h.2.mlp.c_fc
h.3.mlp.c_fc
h.4.mlp.c_fc
h.5.mlp.c_fc
h.6.mlp.c_fc
h.7.mlp.c_fc
h.0.mlp.c_proj
h.1.mlp.c_proj
h.2.mlp.c_proj
h.3.mlp.c_proj
h.4.mlp.c_proj
h.5.mlp.c_proj
h.6.mlp.c_proj
h.7.mlp.c_proj

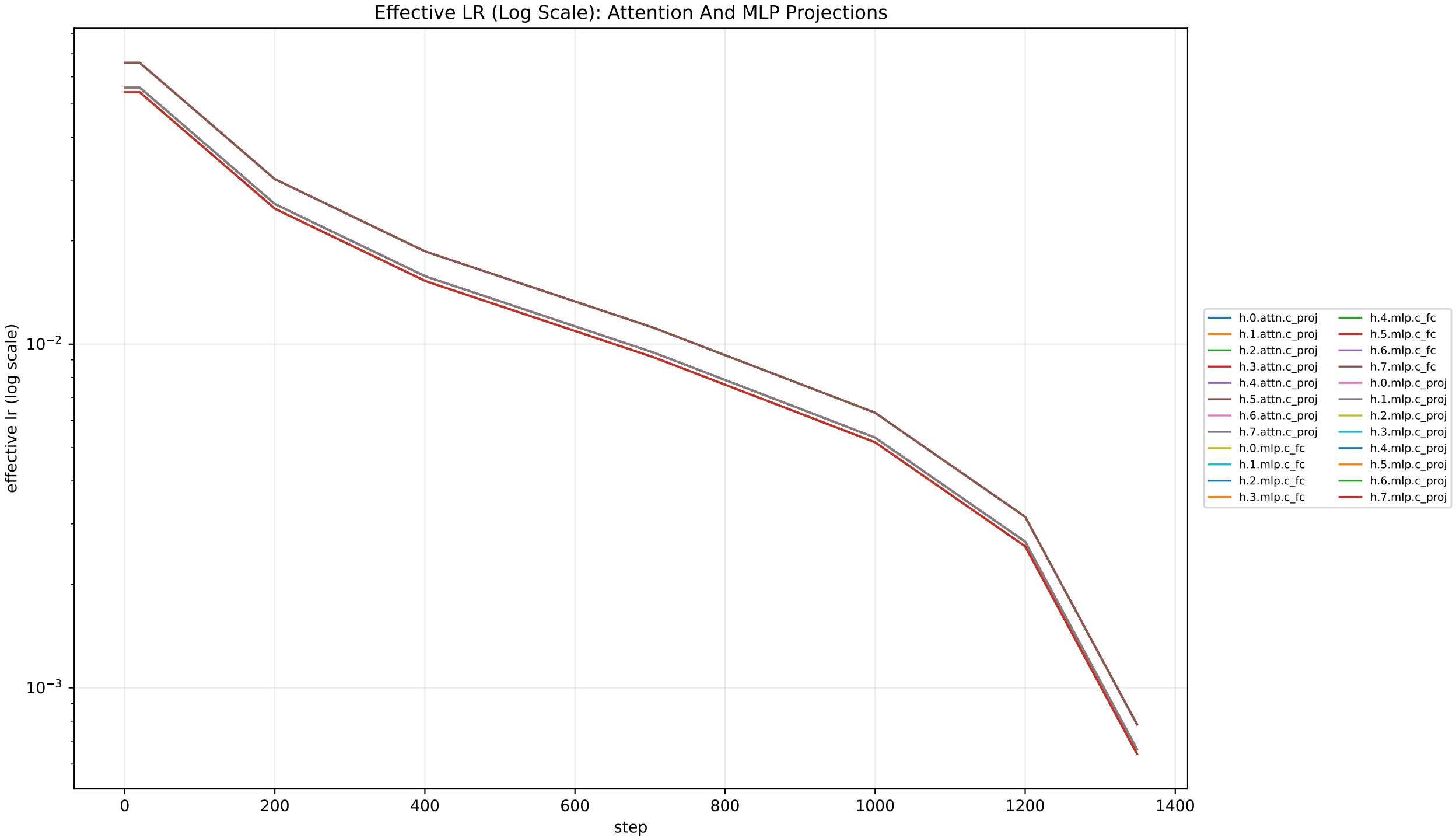

Update Norms: Lambda Scalars
update norms
step
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0
200
400
600
800
1000
1200
1400
h.0.resid_lambdas
h.1.resid_lambdas
h.2.resid_lambdas
h.3.resid_lambdas
h.4.resid_lambdas
h.5.resid_lambdas
h.6.resid_lambdas
h.7.resid_lambdas
h.0.x0_lambdas
h.1.x0_lambdas
h.2.x0_lambdas
h.3.x0_lambdas
h.4.x0_lambdas
h.5.x0_lambdas
h.6.x0_lambdas
h.7.x0_lambdas

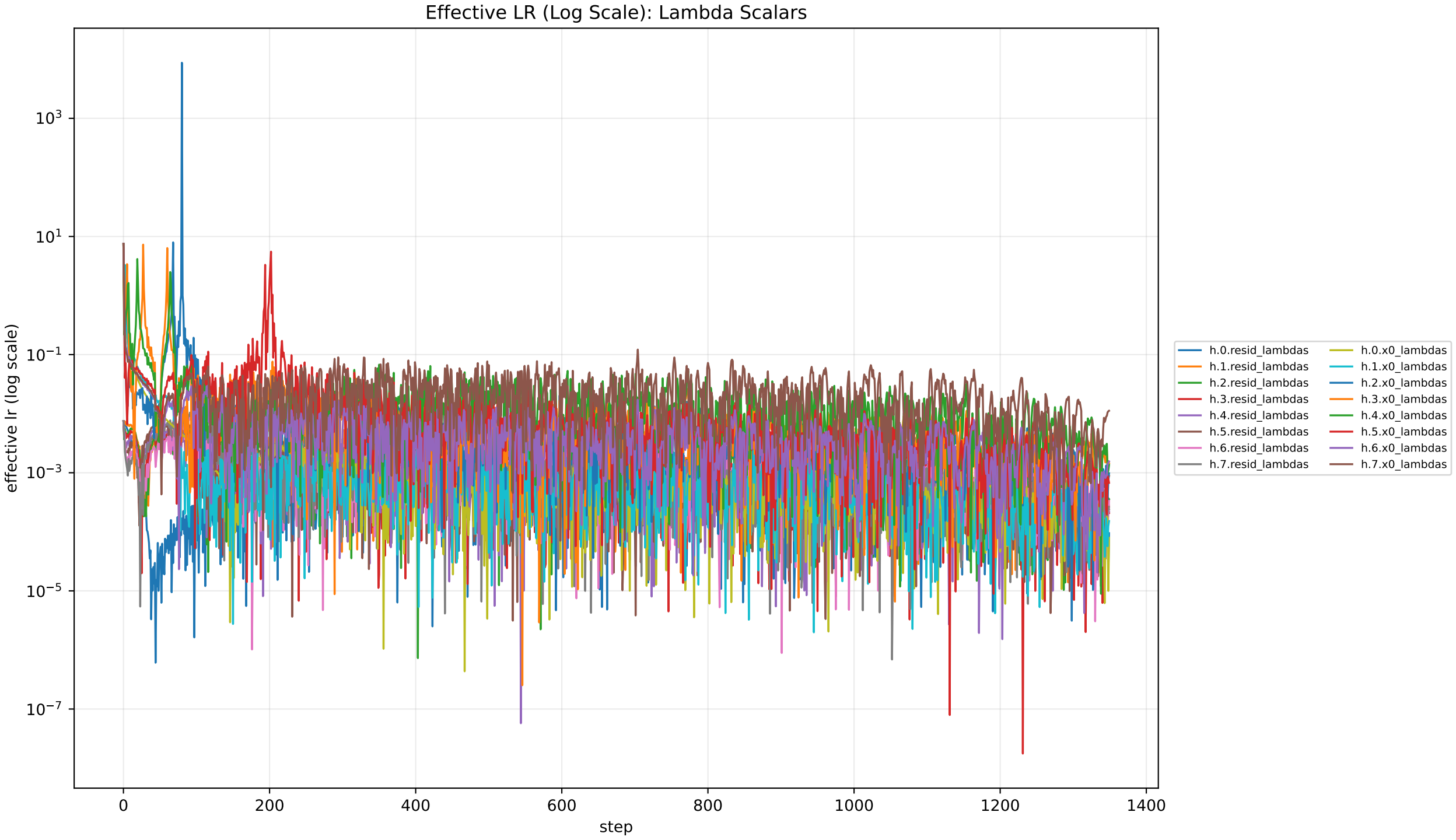

Update Norms: Value Embeddings And Gates
update norms
step
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
0
200
400
600
800
1000
1200
1400
h.1.attn.ve_gate
h.3.attn.ve_gate
h.5.attn.ve_gate
h.7.attn.ve_gate
h.1.ve
h.3.ve
h.5.ve
h.7.ve

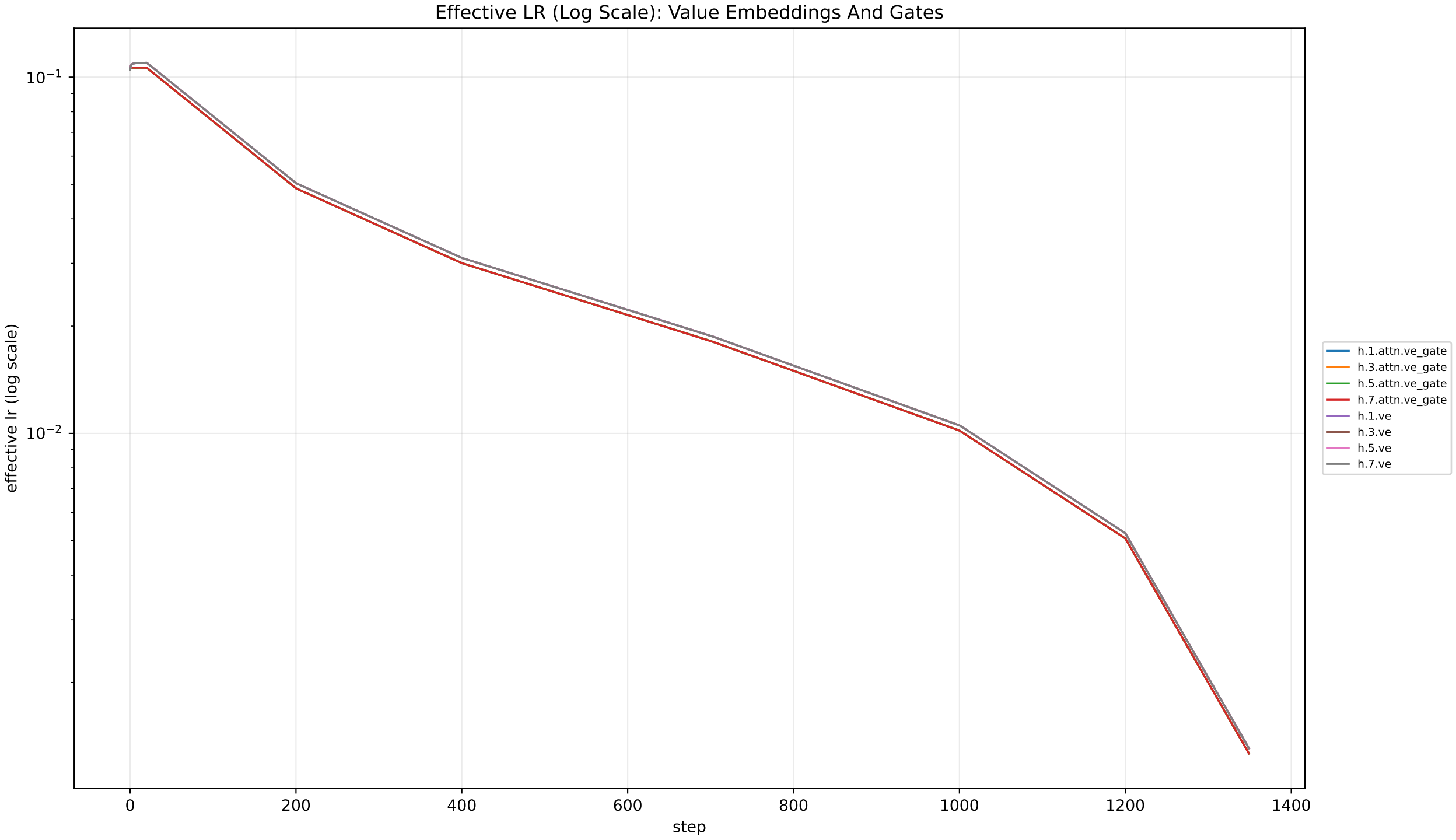

Update Norms (Step >= 20): Lambda Scalars
update norms (step >= 20)
step
0.0
0.1
0.2
0.3
0.4
0
200
400
600
800
1000
1200
1400
h.0.resid_lambdas
h.1.resid_lambdas
h.2.resid_lambdas
h.3.resid_lambdas
h.4.resid_lambdas
h.5.resid_lambdas
h.6.resid_lambdas
h.7.resid_lambdas
h.0.x0_lambdas
h.1.x0_lambdas
h.2.x0_lambdas
h.3.x0_lambdas
h.4.x0_lambdas
h.5.x0_lambdas
h.6.x0_lambdas
h.7.x0_lambdas

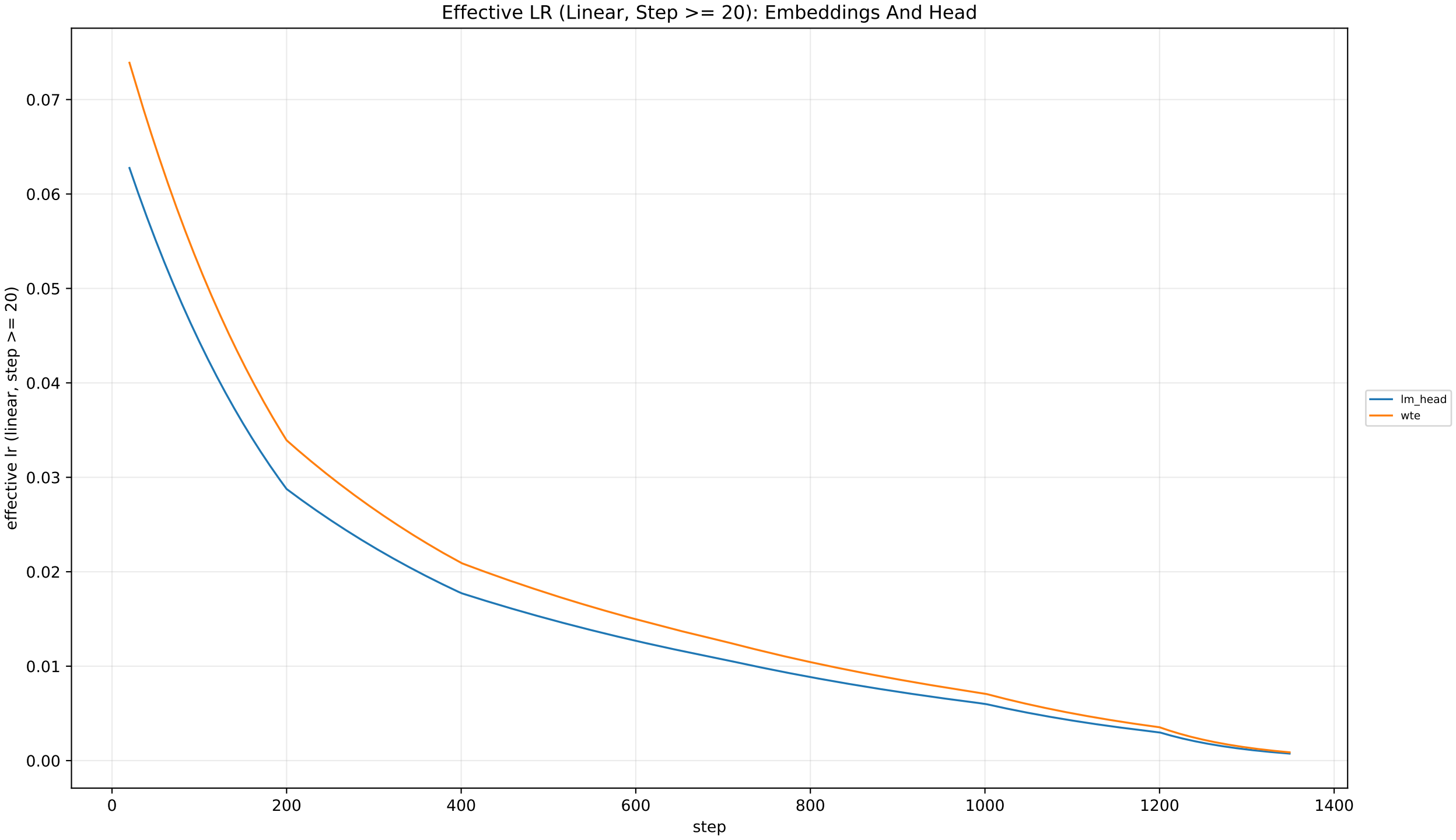

Effective LR (Log Scale): Embeddings And Head
effective lr (log scale)
10^{-2}
10^{-3}
0
200
400
600
800
1000
1200
1400
step
lm_head
wte

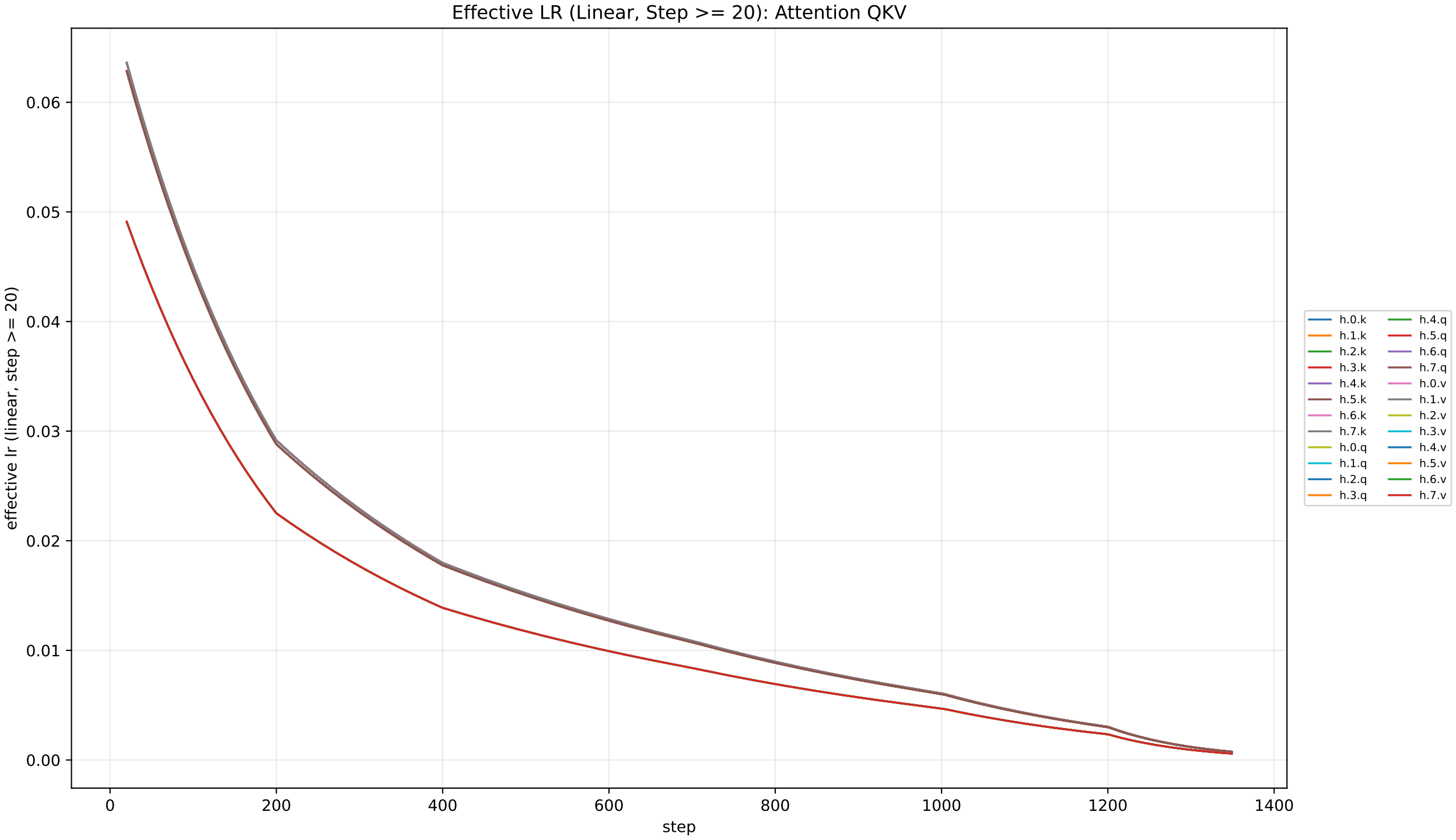

Effective LR (Log Scale): Attention QKV
effective lr (log scale)
10^{-2}
10^{-3}
0
200
400
600
800
1000
1200
1400
step
h.0.k
h.1.k
h.2.k
h.3.k
h.4.k
h.5.k
h.6.k
h.7.k
h.0.q
h.1.q
h.2.q
h.3.q
h.4.q
h.5.q
h.6.q
h.7.q
h.0.v
h.1.v
h.2.v
h.3.v
h.4.v
h.5.v
h.6.v
h.7.v

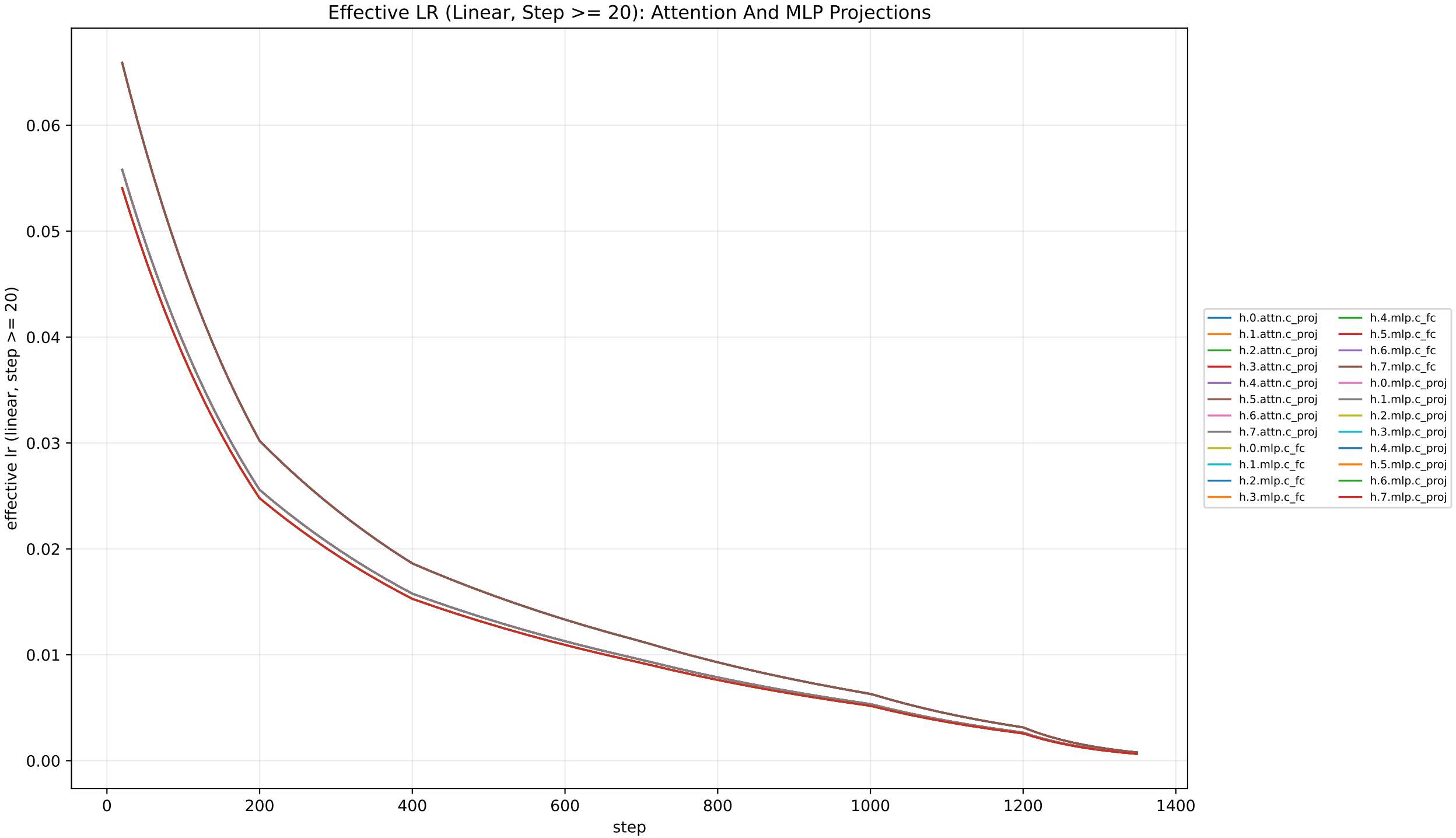

Effective LR (Log Scale): Attention And MLP Projections
effective lr (log scale)
10^{-2}
10^{-3}
0
200
400
600
800
1000
1200
1400
step
h.0.attn.c_proj
h.1.attn.c_proj
h.2.attn.c_proj
h.3.attn.c_proj
h.4.attn.c_proj
h.5.attn.c_proj
h.6.attn.c_proj
h.7.attn.c_proj
h.0.mlp.c_fc
h.1.mlp.c_fc
h.2.mlp.c_fc
h.3.mlp.c_fc
h.4.mlp.c_fc
h.5.mlp.c_fc
h.6.mlp.c_fc
h.7.mlp.c_fc
h.0.mlp.c_proj
h.1.mlp.c_proj
h.2.mlp.c_proj
h.3.mlp.c_proj
h.4.mlp.c_proj
h.5.mlp.c_proj
h.6.mlp.c_proj
h.7.mlp.c_proj

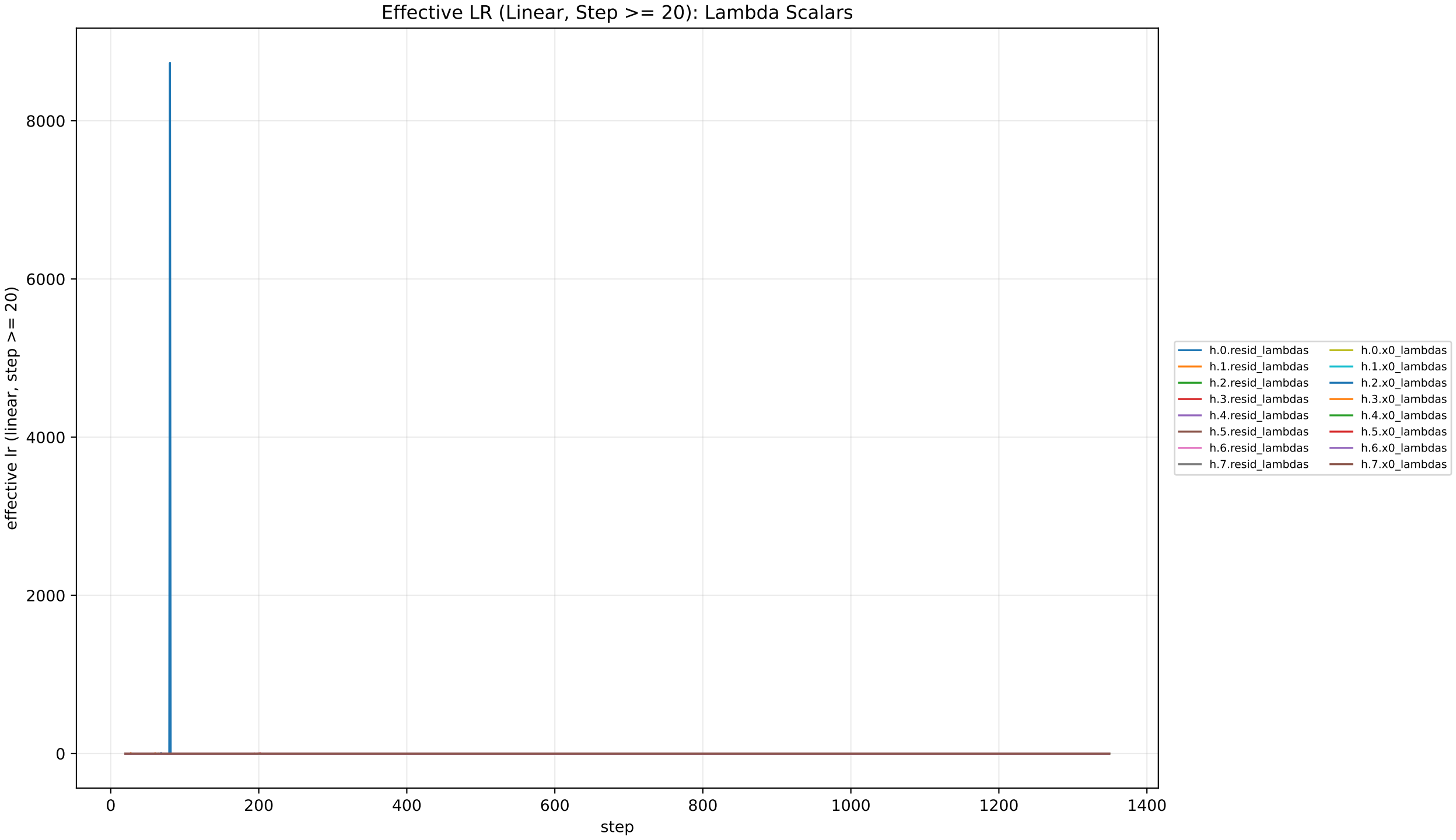

Effective LR (Log Scale): Lambda Scalars
effective lr (log scale)
step
10^{3}
10^{1}
10^{-1}
10^{-3}
10^{-5}
10^{-7}
0
200
400
600
800
1000
1200
1400
h.0.resid_lambdas
h.1.resid_lambdas
h.2.resid_lambdas
h.3.resid_lambdas
h.4.resid_lambdas
h.5.resid_lambdas
h.6.resid_lambdas
h.7.resid_lambdas
h.0.x0_lambdas
h.1.x0_lambdas
h.2.x0_lambdas
h.3.x0_lambdas
h.4.x0_lambdas
h.5.x0_lambdas
h.6.x0_lambdas
h.7.x0_lambdas

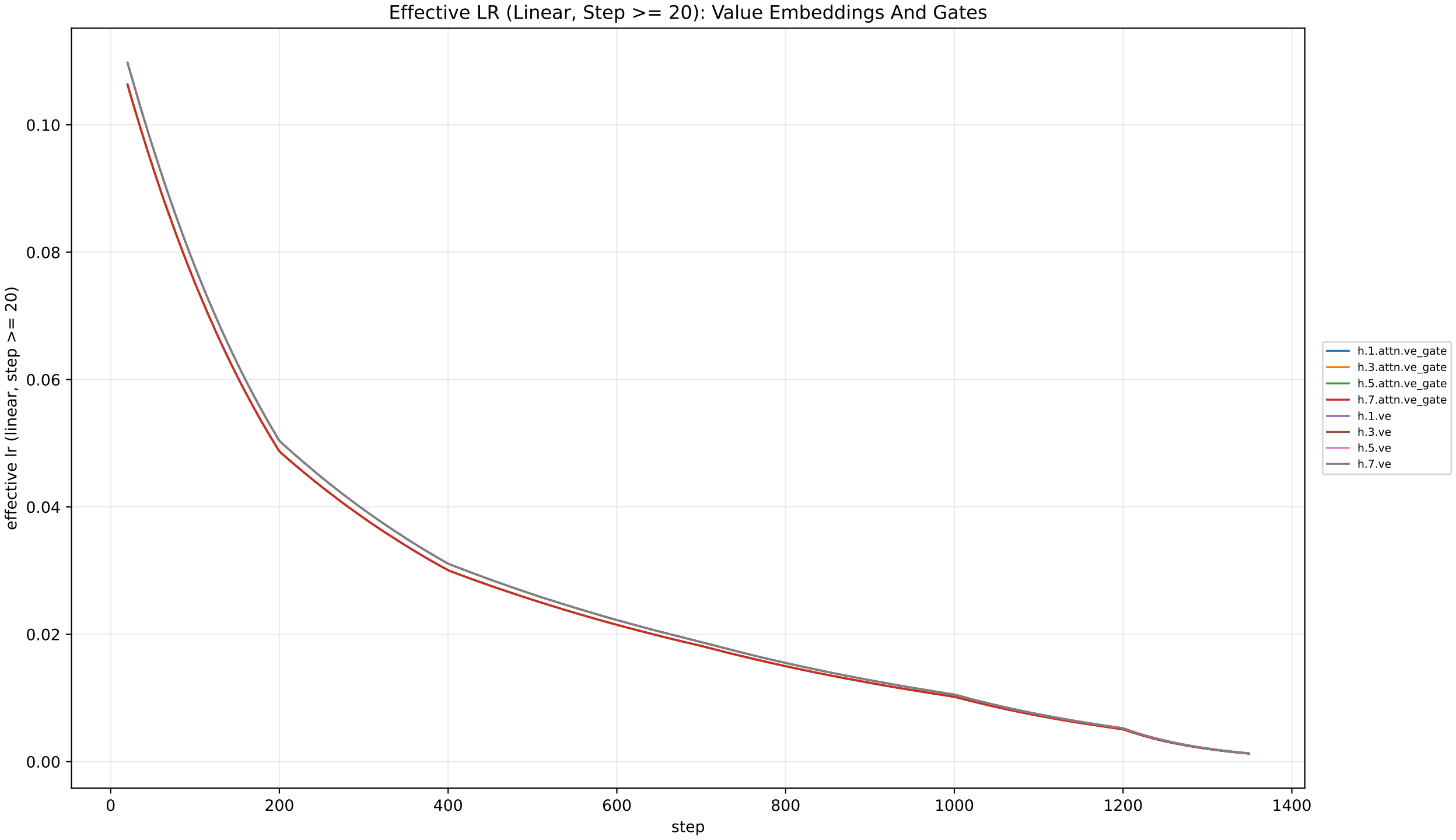

Effective LR (Log Scale): Value Embeddings And Gates
effective lr (log scale)
10^{-1}
10^{-2}
0
200
400
600
800
1000
1200
1400
step
h.1.attn.ve_gate
h.3.attn.ve_gate
h.5.attn.ve_gate
h.7.attn.ve_gate
h.1.ve
h.3.ve
h.5.ve
h.7.ve


Effective LR (Linear, Step >= 20): Embeddings And Head
effective lr (linear, step >= 20)
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
0
200
400
600
800
1000
1200
1400
step
lm_head
wte


Effective LR (Linear, Step >= 20): Attention QKV
effective lr (linear, step >= 20)
0.06
0.05
0.04
0.03
0.02
0.01
0.00
0
200
400
600
800
1000
1200
1400
step
h.0.k
h.1.k
h.2.k
h.3.k
h.4.k
h.5.k
h.6.k
h.7.k
h.0.q
h.1.q
h.2.q
h.3.q
h.4.q
h.5.q
h.6.q
h.7.q
h.0.v
h.1.v
h.2.v
h.3.v
h.4.v
h.5.v
h.6.v
h.7.v


Effective LR (Linear, Step >= 20): Attention And MLP Projections
effective lr (linear, step >= 20)
0.06
0.05
0.04
0.03
0.02
0.01
0.00
0
200
400
600
800
1000
1200
1400
step
h.0.attn.c_proj
h.1.attn.c_proj
h.2.attn.c_proj
h.3.attn.c_proj
h.4.attn.c_proj
h.5.attn.c_proj
h.6.attn.c_proj
h.7.attn.c_proj
h.0.mlp.c_fc
h.1.mlp.c_fc
h.2.mlp.c_fc
h.3.mlp.c_fc
h.4.mlp.c_fc
h.5.mlp.c_fc
h.6.mlp.c_fc
h.7.mlp.c_fc
h.0.mlp.c_proj
h.1.mlp.c_proj
h.2.mlp.c_proj
h.3.mlp.c_proj
h.4.mlp.c_proj
h.5.mlp.c_proj
h.6.mlp.c_proj
h.7.mlp.c_proj


Effective LR (Linear, Step >= 20): Lambda Scalars
effective lr (linear, step >= 20)
8000
6000
4000
2000
0
0
200
400
600
800
1000
1200
1400
step
h.0.resid_lambdas
h.1.resid_lambdas
h.2.resid_lambdas
h.3.resid_lambdas
h.4.resid_lambdas
h.5.resid_lambdas
h.6.resid_lambdas
h.7.resid_lambdas
h.0.x0_lambdas
h.1.x0_lambdas
h.2.x0_lambdas
h.3.x0_lambdas
h.4.x0_lambdas
h.5.x0_lambdas
h.6.x0_lambdas
h.7.x0_lambdas


Effective LR (Linear, Step >= 20): Value Embeddings And Gates
effective lr (linear, step >= 20)
0.10
0.08
0.06
0.04
0.02
0.00
0
200
400
600
800
1000
1200
1400
step
h.1.attn.ve_gate
h.3.attn.ve_gate
h.5.attn.ve_gate
h.7.attn.ve_gate
h.1.ve
h.3.ve
h.5.ve
h.7.ve